Technical Service Manual

Firmware, Processor System, Remote Command Set and Calibration Data

Reverse-Engineering Supplement to the 34420A Service Guide
Document 34420-90T01
Derived from ROM images 34420-88851, 34420-88871 and 34420-88872
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Scope

This manual documents the internals of the 34420A Nano Volt / Micro Ohm Meter: the processor and memory system of the floating main controller, the organisation of its firmware, the data structures of its SCPI parser, the complete remote command set including the unpublished commands, the calibration and non-volatile data subsystem, and the diagnostic and error reporting.

It supplements the manufacturer's documentation and does not replace it. Operation, performance verification and the specifications are in the User's Guide and the Service Guide, which remain the controlling documents for anything a customer may do to the instrument, and the companion 34420A User's Manual gathers them into one volume. Where a fact is already published this manual cites the page; it prints in full only what those documents do not carry and where the firmware disagrees with them.

Provenance

The firmware material comes from disassembly of the three main-processor ROM images: 34420-88851 (revision 5.0), 34420-88871 (revision 9.0) and 34420-88872 (revision 10.0), using an MCS-96 disassembler written for the purpose. Recursive traversal from the reset and interrupt vectors, the four bank-transfer vector tables and eight indirect-branch dispatch tables found by hand covers 76, 46, 39 and 38 percent of the code space of banks 0 to 3; the remainder is tables and text. The hardware material comes from the 34420A schematic set (component locators 34420-66501 and 34420-66502 and eleven sheets) and the Service Guide's self-test descriptions.

A 34420A carrying main firmware revision 10.0 was on the GPIB throughout, and every command path recovered from the ROM was tested against it. Chapter 10 records the results and how to repeat the tests. No command that writes calibration memory was sent to it.

Applicability

The command set, the parser tables and the error-message table are byte-for-byte identical in main firmware revisions 5.0, 9.0 and 10.0; Chapter 3 lists the differences. Addresses are those of revision 10.0 unless stated otherwise, and tables that vary by revision give all three. The front-panel processor (revision 5.0 in the instrument examined) and the earth-referenced I/O processor (revision 2.0) run from internal program memory that was not available; what is said about them is inferred from the main processor's side of their links and from the schematics.

Conventions

Hexadecimal quantities carry a $ prefix: $2080 is an address, $C0 a byte value; decimal is unmarked. Bits are numbered from zero at the least significant end. 80C196 register-file addresses are written R<nn>h and special-function registers by name. Machine text (commands, responses, mnemonics and register names) is set in the monospaced face. In command syntax, angle brackets enclose a value the user supplies, square brackets an optional element and braces a choice. SCPI mnemonics show the short form in upper case and the rest of the long form in lower case.

The manufacturer's documents are cited by short title: *User's Guide* (34420-90001, edition 8), *Service Guide* (34420-90010, edition 5), *User's Manual* (the first edition of the User's Guide, published under that title) and *Manual Update* (34420-90091). Findings carry the evidence behind them: 'confirmed on the instrument' means tested on hardware, and 'from firmware analysis alone' means resting only on the disassembly. Chapter 10 collects both.

Preparation

Prepared by Jared Cabot, with the firmware analysis and text prepared with Claude (Anthropic). The disassembly, its interpretation, the instrument tests and the ROM-derived tables are this project's own work; published figures, page references and the published status of each command come from the four manufacturer documents above.

Disclaimer

This is not a publication of Hewlett-Packard, Agilent Technologies or Keysight Technologies and is not endorsed by them. Trade names are used for identification only. The floating measurement section may sit at a hazardous potential with respect to earth. The diagnostic facilities described in Chapters 6 and 7 write to the instrument's memory and calibration data, and used carelessly they will destroy its calibration.

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1

General Information

Instrument Description

The 34420A is a 7½-digit two-channel nanovoltmeter and micro-ohmmeter. Its measuring capabilities, ranges and specifications are described in the User's Guide, pages 10 and 34 to 37, and are not repeated here.

The feature of the instrument that governs everything in this manual is its electrical division. It is split into a floating measurement section and an earth-referenced interface section, separated by an optical isolation barrier, and each section has its own processor. A third processor drives the front panel and runs from a supply referenced to -18 V so that it can drive the vacuum-fluorescent display directly. All three report their firmware revisions jointly in the response to `*IDN?`, which is the subject of the last section of this chapter.

Assemblies and Processors

The instrument is built on two printed-circuit assemblies. 34420-66501 carries the measurement circuitry, both main processors and the power supplies; 34420-66502 is the front-panel assembly. The processors and the major digital devices are listed below, with the schematic sheet on which each appears.

Table 1-1. Processors and Major Digital Devices

Ref	Device	Function	Sheet
U500	Intel 80C196KB (see Chapter 2)	Floating main controller, 12 MHz	Floating Logic
U501	Custom gate array	Bus demultiplexer, chip selects, serial links, A/D control	Floating Logic
U502 / J502	27C020 (256K x 8)	Main firmware EPROM	Floating Logic
U503	M5M5256BFP-70L	Main static RAM, 32K x 8 device	Floating Logic
U505	NMC93C66	Calibration EEPROM, 256 x 16 Microwire	Floating Logic
U506, U704	HCPL-2211	Isolation-barrier optocouplers	Floating / Earth Logic
U601-U606	74HC4094 x 6	Configuration data shift registers, 48 of 72 bits	Config Data Shift Reg
U700	Intel 87C51FA	Earth-referenced I/O processor, 12 MHz	Earth Referenced Logic
U701	TMS9914A	GPIB (IEEE 488) controller	Earth Referenced Logic
U702 / U703	75ALS160 / 75ALS162	GPIB bus transceivers	Earth Referenced Logic
U706	LT1180	RS-232 transceiver	Earth Referenced Logic
U602	Intel 87C51	Front-panel processor	Display and Keyboard
U603	ULS75518	Vacuum-fluorescent display driver	Display and Keyboard
U606	34420-89301	Vacuum-fluorescent display	Display and Keyboard

Note



The reference designators for the configuration shift registers (U601-U606) fall in the same numeric block as the front-panel designators but belong to the main assembly. They appear on the Configuration Data Shift Registers sheet, not on the Display and Keyboard sheet.

Firmware Revisions

Three main-processor ROM images were available for this work. All are 262144 bytes and program a 27C020 or equivalent. The first field of the revision string reported by `*IDN?` is taken from the first two bytes of each bank header, described in Chapter 3; the other two fields come from the other two processors, as the last section of this chapter shows.

Table 1-2. Main-Processor ROM Images Examined

Part number	Revision	Size	MD5
34420-88851	5.0	262144 bytes	`f7bf93bdec4e7eca80ad9b0d2f389ed3`
34420-88871	9.0	262144 bytes	`62d294e479a2ddec82dc22a2d351c43c`
34420-88872	10.0	262144 bytes	`c4fad3f5dfdbadf188d43f39d9e2eb41`

The badge on the front panel changes across this range (the revision 5.0 part is an HP number, revision 9.0 an Agilent number and revision 10.0 a Keysight number) but the command set does not. Chapter 3 sets out what does differ.

Reading the Revision Codes

The User's Guide, page 115, gives the field order of the `*IDN?` revision string as measurement processor, input/output processor, front-panel processor, and directs the reader to the front-panel `11: REVISION` item for the same three numbers. This section does not restate that. It records where in the firmware each of the three numbers comes from, because the source is not evident from the published text.

The formatter is at bank 3 $3E52. It writes six unsigned values through the format string `%u.%u-%u.%u-%u.%u` at $F294, and the result is substituted into `HEWLETT-PACKARD,34420A,0,%s` at $F2A6 by the routine at $3ED4. Arguments are pushed right to left, so the push order in the listing is the reverse of the printed order.

Table 1-3. Origin of the Three Revision Fields

Field	Read from	Written by	Processor
1	`$B472` / `$B473`	ROM constant: the first two bytes of the bank header	Measurement, U500
2	`$17EA`, printed as `n/10` . `n%10`	`$3070`, inside the serial receive interrupt `INT09` at `$2032` -> `$3030`, from `SBUF_RX`	Input/output, U700
3	`$14EC`, printed as `n.0`	bank 1 `$AFE0`, fed from `$14A6`, which only `$2283` writes, reading the gate-array link register `$1FF9`	Front panel, U602

Each of the two RAM cells has exactly one writer in the whole 256 KB image, and the two writers sit on demonstrably different hardware: field 2 is filled from the on-chip UART, which is the optically isolated link to U700, and field 3 from the $1FF8-$1FFB gate-array window, which is the front-panel link. The published order is therefore correct. The same push sequence appears in revisions 5.0 and 9.0, so this is not an artefact of one build.

One question remains open, from firmware analysis alone. The byte the front panel returns is passed through the transform `v' = ((v & 7) - 1) + 6 * ((v >> 3) - 1)` at $AFED, with a further `v' -= 2` when the result exceeds $0F. That is the shape of a row-and-column decode, not of a revision number, so field 3 may be better described as a front-panel identification code that doubles as a revision. Resolving it needs the U602 program memory, which was not available.

The manufacturer's name in the identification string is fixed in ROM and does not change with the badge: it is still `HEWLETT-PACKARD` in the Keysight-numbered revision 10.0 image.

2

Processor and Memory System

The Main Controller

U500 is an Intel 80C196KB in a 68-pin plastic leaded chip carrier. It runs from a 12 MHz oscillator module (U405) applied to XTAL1. EA* is tied to ground, so the device executes entirely from external memory; BUSWIDTH is pulled to ground through R508, selecting the 8-bit external bus; READY is pulled high through R512, so no wait states are inserted. NMI and CDE are grounded. The reference input for the on-chip 10-bit analogue-to-digital converter is driven from the +5REF supply, and that converter is used for housekeeping measurements rather than for the measurement path.

The chip configuration byte at $2018 is $0C in all four banks of all three ROM images.

The schematic labels U500 80C196KB#1 and this manual follows it, but the firmware uses the window select register WSR at SFR $14 (the sequence DI, LDB WSR,#0Fh, read TIMER2, EI, CLRB WSR at $2643 in the resident region), and WSR is an 80C196KC feature that the KB does not have. Either the schematic predates a running change to the KC, or the part fitted is a KC in a position the schematic still calls a KB, or the sequence is harmless dead code on a KB. The evidence does not choose between them and nothing else in this manual depends on which part is fitted. Settling it requires the part marking on U500 of a real board.

Address Space

The 80C196 has a single 64 KB address space. On this instrument it is divided as follows. The gate array U501 generates the two chip selects RAMCE* and ROMCE* from the latched address.

Table 2-1. Main Processor Address Space

Address	Contents	Device
$0000-$00FF	80C196 register file and special-function registers	on chip
$0100-$1FF7	External static RAM, one 8 KB page selected by RAMA13	U503
$1FF8-$1FFF	Gate-array registers (front-panel link and status)	U501
$2000-$203F	Interrupt vectors, chip configuration byte, security key	EPROM
$2080-$FFFF	Program and constant data, one 64 KB bank of four	U502

The stack pointer is initialised to $1FF8 and grows downward through the RAM page. A second, software-managed stack used only by the bank-transfer mechanism is anchored at $17F8 and grows upward; see Chapter 3.

Note



Addresses $1FF8 to $1FFB lie inside the RAM window but are decoded to the gate array, not to the RAM. From firmware analysis alone: reading $1FF8 returns a status byte whose bit 4 is polled by the front-panel handler, writing $1FF8 loads a control byte, $1FF9 is the front-panel data register, and writing $1FFB acknowledges the interrupt. The schematic does not show the register assignment.

Program Memory Banking

The EPROM is a 256 KB device with eighteen address lines. The processor supplies A0 to A15. The remaining two, A16 and A17, are driven directly from port 1 bits 6 and 7 of U500 (pins 31 and 32, whose `HLDA*` and `HOLD*` alternate functions are unused here). The EPROM therefore presents one of four 64 KB banks at a time across the whole of the processor's program space, and a bank switch changes every address above $2000 at once.

Table 2-2. ROM Bank Selection

PORT1 bits 7:6	A17:A16	ROM offset	Contents
00	00	`$00000`	Bank 0: measurement and calibration
01	01	`$10000`	Bank 1: front panel and display
10	10	`$20000`	Bank 2: message text and language emulation
11	11	`$30000`	Bank 3: parser and executive (boot bank)

Because a bank switch is total, the code that performs it must exist at the same address in every bank. That requirement shapes the whole firmware and is the subject of Chapter 3.

Note



Bit 0 of PORT1 is the front-panel interrupt line FPINT and is manipulated by the display handler, while bits 6 and 7 select the ROM bank. Any code that writes PORT1 must preserve the bits it does not own; the firmware keeps a shadow copy in R48h for this reason.

Data Memory Paging

U503 is a 32K x 8 static RAM. Its A0 to A12 inputs come from the processor address bus, giving an 8 KB window; A14 is tied to +5 V, so only the upper half of the device is reachable; and A13 is driven by `RAMA13`, port 2 bit 6 of U500 (pin 33, the `T2UP-DN` alternate function). Sixteen kilobytes of the device are therefore usable as two 8 KB pages sharing one window.

Note



Every RAM address quoted in this manual is a window address in the range $0100-$1FFF. Which physical page it lands in depends on the state of RAMA13 at the time, and the firmware's page discipline was not established. The addresses used by the resident code (the bank-transfer stack at $17F8, the crash buffer at $1488 and the front-panel state at $14A6 and $14AC) are all in the low half of the window and appear to be page-independent in practice.

The Gate Array U501

U501 is a custom device in a 68-pin package. It sits between the processor and everything else on the floating side. Its functions, read from the Floating Logic sheet, are:

- Latching the multiplexed low address from `ADDAT(7:0)` under `ALE` and presenting `A(7:0)` to the EPROM and RAM.
- Decoding `ROMCE*` and `RAMCE*`.
- Running the synchronous serial link to the front-panel processor: `FPDO`, `FPDI`, `FPSK*`, with `INTR` brought out as `CHINT`.
- Running the configuration shift-register link: `SERCK`, `SERDAT`, `SERRBK`.
- Generating the A/D control signals `ADSW` and `SYNC` and returning the comparator state `COMP`.
- Providing the counter output `CNTROUT` used for the voltmeter-complete signal.

The gate array's interrupt output `CHINT` is wired to P0.7 of U500, which the 80C196 can use as an alternate `EXTINT` source. The corresponding vector INT07 at $200E holds a real handler, at $21AC, while INT06, the on-chip serial port's combined vector, is left pointing at the unexpected-interrupt trap. The instrument uses the separate transmit and receive vectors instead.

Port Assignments

Table 2-3. U500 Port and Special-Function Pin Use

Pin	Signal name	Function
P0.0 / ACH0	FLASH	Analogue housekeeping input from the function-switching sheet
P0.1 / ACH1	JM500 sense	Calibration security jumper, pulled up by R504
P0.2-P0.6	-	Grounded
P0.7 / ACH7	CHINT	Gate-array interrupt, used as EXTINT
P1.0	FPINT	Front-panel attention output
P1.1	FPRST	Front-panel reset output
P1.2-P1.4	AZ2, AZ1, AZ0	Autozero / input-switch control to the analogue section
P1.5	-	Unused
P1.6	A(16)	EPROM address line 16, ROM bank bit 0
P1.7	A(17)	EPROM address line 17, ROM bank bit 1
P2.0 / TXD	OP_TXD	Isolated serial link to the I/O processor
P2.1 / RXD	OP_RXD	Isolated serial link from the I/O processor
P2.2 / EXTINT	-	Grounded
P2.3 / T2CLK	CNTROUT	Counter input from the gate array
P2.4 / T2RST	CALDO	Calibration EEPROM data out (read by the processor)
P2.5 / PWM	-	Unused
P2.6 / T2UP-DN	RAMA13	Static RAM address line 13, RAM page select
P2.7 / T2CAP	SYNC	A/D synchronisation
HSI.0	LSENSE	Line-frequency sense input
HSO.0	XRLYRST	Latching relay reset
HSO.1	CALCS	Calibration EEPROM chip select
HSO.2	CALSK	Calibration EEPROM clock
HSO.3	CALDI	Calibration EEPROM data in

These assignments are read from the Floating Logic sheet, and the firmware's use of them is consistent: the reset code writes $C0 to PORT1 and $BF to PORT2 before anything else, selecting bank 3 with RAMA13 low, and the front-panel handler changes only bit 0 of its PORT1 shadow.

Calibration Memory

U505 is an NMC93C66, a 4096-bit Microwire serial EEPROM organised as 256 sixteen-bit words. Its chip select, clock and data-in lines are driven from high-speed output channels 1, 2 and 3 of U500, so the processor clocks the device from the HSO unit rather than by bit-banging a port. Data returns on CALDO to P2.4, pulled up by R502.

These 512 bytes hold every constant the instrument does not recompute at power-up: the calibration corrections for each function and range, the calibration count and message, the security state and code, the GPIB address and the stored interface settings. Chapter 7 covers the organisation.

3

Firmware Organisation

ROM Image Layout

The ROM image is four 64 KB banks laid end to end. Within each bank the layout is the same, and it is the layout an 80C196 requires: the interrupt vectors at the bottom of the external program space, the chip configuration byte at $2018, and the reset entry at $2080.

Table 3-1. Layout Within One 64 KB Bank

Address	Contents
$0000-$1FFF	Not fetched. The processor sees RAM here. In the images this region is $FF filler except for a small block of $00/$01 bytes at $0100-$061F in banks of revisions 9.0 and 10.0.
$2000-$2013	Interrupt vectors INT00 to INT07, the TRAP vector and the unimplemented-opcode vector
$2018	Chip configuration byte, $0C
$2020-$202F	Security key area, erased ($FF)
$2030-$203F	Upper interrupt vectors INT08 to INT14 and NMI
$2080	Reset entry
$2080-$35B3	Resident code, identical in all four banks apart from three patched words
$35B4-	Bank-private code and data
-$FF9D	End of used space; the value differs per bank and revision
$FF9E-$FFFD	A table common to all banks and revisions, function not established

The last used address in each bank of each image is:

Table 3-2. Extent of Used Space by Bank

Revision	Bank 0	Bank 1	Bank 2	Bank 3
5.0	$FDD3	$FAA5	$C523	$FABB
9.0	$FDDB	$FA71	$C524	$FAC3
10.0	$FDDB	$FA85	$C524	$FAC3

The Bank Header

Each bank carries a seventy-byte header immediately below its bank-transfer vector table, at the table's base address less $46. The first byte of that header is the firmware major revision in binary and the second is the minor revision: $0A and $00 in every bank of the revision 10.0 image, $09 and $00 in revision 9.0, $05 and $00 in revision 5.0. They are formatted through `%u.%u-%u.%u-%u.%u` at bank 0 $F294 into `HEWLETT-PACKARD,34420A,0,%s` at $F2A6, which is the `*IDN?` response, and they drive the front-panel `REVISION` item.

The remaining sixty-eight bytes are not one bitmap. They are several ten-element arrays of $00 and $01 bytes, one element per measurement function, at offsets +1, +11, +22, +31, +41 and +51. They are per-function property flags, replicated as read-only data in each bank. Which property each array carries was not established. The whole block is identical in every bank of every revision examined apart from the two revision bytes.

Table 3-3. Bank Header and Vector Table Addresses

Revision	Bank	Header	Vector table	Revision byte
5.0	0	$EC74	$ECBA	$05
	1	$D036	$D07C	$05
	2	$83CA	$8410	$05
	3	$B46A	$B4B0	$05
9.0	0	$EC7C	$ECC2	$09
	1	$D002	$D048	$09
	2	$83C4	$840A	$09
	3	$B472	$B4B8	$09
10.0	0	$EC7C	$ECC2	$0A
	1	$D016	$D05C	$0A
	2	$83C4	$840A	$0A
	3	$B472	$B4B8	$0A

Reset and Initialisation

Execution begins at $2080 in whichever bank the port pins happen to select at reset. Because the resident region is duplicated, that does not matter: the first instructions of the reset sequence set the bank explicitly. The sequence, from the revision 10.0 image, is:

```
2080  CLR   R4Eh               ; clear flag word
2082  LD    R50h, #FADFh
2086  CLRB  R54h
2088  LDBSE RA2h, #01h
208B  LD    SP,   #1FF8h       ; hardware stack top
208F  LDB   R48h, #C0h         ; PORT1 shadow: bank 3, FPINT high
2092  STB   R48h, PORT1         ; select ROM bank 3
2095  LDB   R49h, #BFh
2098  STB   R49h, PORT2         ; RAMA13 low
209B  LD    RB4h, #17F8h       ; bank-transfer stack pointer
209F  ...                      ; A/D, serial and timer registers
20C2  LDB   R4Ah, #40h
20C5  STB   R4Ah, 1FF8h        ; gate-array control
20CA  STB   ZERO_REG, 1FFBh    ; clear pending front-panel interrupt
20CF  STB   ZERO_REG, 14ECh
20D4  LCALL <bank init>        ; patched per bank
20D7  LCALL 2C50h              ; resident initialisation
20DA  ...
20DF  LDB   R4Dh, #25h
20E2  LDB   INT_MASK,  R4Dh    ; enable INT00, INT02, INT05
20E5  LDB   R4Ch, #00h
20E8  LDB   INT_MASK1, R4Ch
20EB  LCALL <bank main>        ; patched per bank
20EE  SJMP  20EBh              ; forever
```

The instrument therefore boots into bank 3, which holds the executive and the command parser. `INT_MASK` is loaded with $25, enabling the timer-overflow, HSI-data-available and software-timer interrupts; the second mask is cleared and the serial interrupts are enabled later by the routine at $20F1.

The Main Loop

The whole of the instrument's foreground work is the two instructions at $20EB and $20EE: an indefinite call to a routine whose address is patched into the resident code per bank. In bank 3 that routine is at $3661; in bank 1 at $399C; in banks 0 and 2 the address resolves to a bare `RET` at $35B4, because those banks are libraries and are never the resident bank.

Table 3-4. Patched Words in the Resident Region

Offset	Instruction	What it selects
`$20D5-$20D6`	`LCALL`	Displacement to the bank's initialisation entry
`$20EC-$20ED`	`LCALL`	Displacement to the bank's main-loop entry
`$2164-$2165`	`LD RB2h,[RB2h]`	Base address of the bank's inter-bank vector table

These three words are the only bytes below $35B4 that differ between banks of one image. Everything else in the resident region is byte-for-byte identical, which is what makes a bank switch safe at any point in it.

Table 3-5. Per-Bank Entry Points, Revision 10.0

Bank	Initialisation	Main loop	Vector table
0	`$35B5`	`$35B4`	`$ECC2`
1	`$399D`	`$399C`	`$D05C`
2	`$35B5`	`$35B4`	`$840A`
3	`$39C0`	`$3661`	`$B4B8`

Interrupt Vectors and Handlers

The vector table is identical in all four banks and in all three revisions. Handlers whose address is $2120 or a two-byte stub that calls $2120 are unused; see the next section.

Table 3-6. Interrupt Vectors, All Banks

Vector	Source	80C196 function	Handler	Note
`$2000`	INT00	Timer 1 overflow	`$259A`	
`$2002`	INT01	A/D conversion complete	`$211E`	unexpected-interrupt trap
`$2004`	INT02	HSI data available	`$25E0`	
`$2006`	INT03	High-speed outputs	`$210A`	unexpected-interrupt trap
`$2008`	INT04	HSI.0 pin	`$210C`	unexpected-interrupt trap
`$200A`	INT05	Software timer	`$25AE`	
`$200C`	INT06	Serial port	`$210E`	unexpected-interrupt trap
`$200E`	INT07	EXTINT	`$21AC`	
`$2010`	-	TRAP instruction	`$2110`	unexpected-interrupt trap
`$2012`	-	Unimplemented opcode	`$2112`	unexpected-interrupt trap
`$2030`	INT08	Serial transmit (TI)	`$2E69`	
`$2032`	INT09	Serial receive (RI)	`$3030`	
`$2034`	INT10	HSI FIFO fourth entry	`$2114`	unexpected-interrupt trap
`$2036`	INT11	Timer 2 capture	`$262E`	
`$2038`	INT12	Timer 2 overflow	`$2116`	unexpected-interrupt trap
`$203A`	INT13	EXTINT1	`$25C2`	
`$203C`	INT14	Unimplemented opcode (upper)	`$211A`	unexpected-interrupt trap
`$203E`	NMI	Non-maskable interrupt	`$211C`	unexpected-interrupt trap

The handlers that do real work are:

`$259A`	INT00, timer 1 overflow. Increments the 32-bit software time base held in R3Eh and R40h and calls the resident routine at $24CA. This is the instrument's only free-running clock.
`$25E0`	INT02, HSI data available. The HSI.0 pin carries `LSENSE`, the line-frequency sense signal, so this handler timestamps mains zero crossings. Self-test 625, 'Unable to sense line frequency', reports its failure.
`$25AE`	INT05, software timer. Masks itself down, then makes an inter-bank call to bank 1 entry $E8. The front panel is therefore serviced on a software timer, not in the foreground loop.
`$21AC`	INT07, EXTINT. The gate array's `CHINT` output. This is the front-panel and configuration-link handler; it drives `FPINT` low through bit 0 of the PORT1 shadow, polls bit 4 of $1FF8 until the link is ready, exchanges a byte through $1FF9, and clears the request by writing $1FFB. It schedules a timeout through the HSO unit by loading `HSO_COMMAND` with $1A and `HSO_TIME` with `TIMER1 + $48`.
`$2E69`	INT08, serial transmit. The isolated link to the earth-referenced processor.
`$3030`	INT09, serial receive. Companion to the above.
`$262E`	INT11, timer 2 capture.
`$25C2`	INT13, EXTINT1. Makes an inter-bank call to bank 0 entry $EF and then disables itself by clearing bit 5 of the second interrupt mask.

The bit assignments of `INT_MASK1` were established from the code rather than assumed: the serial initialisation routine at $20F1 sets bit 1 and then bit 0 of its mask shadow immediately after reading `SBUF` and `SP_STAT`, which identifies bit 0 as the transmit interrupt and bit 1 as the receive interrupt.

The Unexpected-Interrupt Trap

Eleven of the eighteen vectors point into a block of two-byte stubs at $210A to $211E, each of which is a short call to $2120. That routine is a crash handler:

```
2120  LD    R1Ch, #1488h        ; buffer in RAM
2124  LDB   R1Eh, #0Fh          ; fifteen words
2127  DI
2128  POP   R20h               ; unwind the stack
212A  ST    R20h, [R1Ch]+      ; and record it
212D  DJNZ  R1Eh, 2128h
2130  RST                      ; reset the processor
```

Fifteen words are lifted off the stack into a buffer at $1488 and the processor is then reset by the `RST` instruction. The buffer survives the reset, because `RST` does not clear RAM. The return address of the stub that was entered is the first word written, so the vector that fired can be recovered from the buffer afterwards.

Note



The crash buffer can be read out over the interface without opening the instrument. `DIAG:PEEK? 1,5256,1` returns the first word at $1488; successive words are two bytes apart. See Chapter 6. The buffer is not cleared at power-up, so a non-zero value may be a record of an event long past.

Inter-Bank Calls

A routine in one bank cannot call a routine in another with an ordinary call, because switching the bank changes the code under the program counter. The firmware solves this with four trampolines in the resident region, one per destination bank, and a vector table in each bank.

The calling sequence is:

PUSH #<entry number>
LCALL 2132h ; destination bank 0
LCALL 2138h ; destination bank 1
LCALL 213Eh ; destination bank 2
LCALL 2144h ; destination bank 3
POP ZERO_REG ; discard the trampoline's return address

Each trampoline loads the bank bits into RB6h and falls into the common body at $2148:

```
2132  DI  / LDB RB6h,#00h / SJMP 2148h     ; bank 0
2138  DI  / LDB RB6h,#40h / SJMP 2148h     ; bank 1
213E  DI  / LDB RB6h,#80h / SJMP 2148h     ; bank 2
2144  DI  / LDB RB6h,#C0h                 ; bank 3


2148  POP   RB2h               ; caller's return address
214A  ST    RB2h, [RB4h]+       ; push it on the transfer stack
214D  ANDB  RB2h, R48h, #C0h    ; current bank bits
2151  ST    RB2h, [RB4h]+       ; push those too
2154  POP   RB2h                ; entry number
2156  ADD   RB2h, RB2h          ; times two
2159  ANDB  R48h, #3Fh
215C  ORB   R48h, RB6h
215F  STB   R48h, PORT1         ; the bank changes here
2162  LD    RB2h, <table>[RB2h] ; fetched from the NEW bank
2167  PUSH  #216Dh              ; return through the far-return stub
216A  EI
216B  BR    [RB2h]              ; enter the routine
```

The instruction at $2162 is what makes the scheme work. Its 16-bit displacement is one of the three words patched per bank, so the operand itself is fetched from the bank that was selected three instructions earlier. Each bank thus indexes its own vector table, and an entry number means whatever the destination bank says it means.

The return path is the stub at $216D:

```
216D  DI
216E  SUB   RB4h, #0002h
2172  ANDB  R48h, #3Fh
2175  ORB   R48h, [RB4h]        ; restore the caller's bank bits
2178  STB   R48h, PORT1
217B  SUB   RB4h, #0002h
217F  LD    RB2h, [RB4h]        ; restore the caller's return address
2182  PUSH  RB2h
2184  EI
2185  BR    [RB2h]
```

Interrupts are disabled across both bank writes. The transfer stack at $17F8 grows upward, two words per level, and is not bounds-checked; a runaway recursion through inter-bank calls would overwrite RAM above it rather than fault.

Table 3-7. Inter-Bank Vector Table Sizes, Revision 10.0

Bank	Table base	Entries to first filler	First live entry number
0	$ECC2	329	$B0
1	$D05C	329	$02
2	$840A	329	$00
3	$B4B8	329	$00

Entries a bank does not export hold the address of a bare RET at $35AE, or zero. A call to an unexported entry number in a bank whose table holds zero at that index transfers to $0000, which is the register file; the firmware does not range-check the entry number.

Differences Between Firmware Revisions

The three images were compared bank by bank. The command tree, the parameter tables and the error-message table were extracted from each and compared as data rather than as bytes.

Revision 9.0 to revision 10.0

Only bank 1, the front panel and display bank, changed. Its code grew by twenty bytes and was relinked, moving its vector table from $D048 to $D05C, and 45345 bytes of the bank differ as a consequence. Every printable string in bank 1 is identical between the two revisions. In banks 0, 2 and 3 exactly one byte changed: the revision byte in the bank header, from $09 to $0A. The command tree, the parameter descriptors and the error table are identical.

From firmware analysis alone, revision 10.0 is a front-panel-only change with no effect on the remote interface. What the change does was not established; no string, table or command is affected by it.

Revision 5.0 to revision 10.0

All four banks differ substantially. The visible changes are:

- The front-panel menu item `1: HPIB ADDR` became `1: GPIB ADDR`, and the interface selection `HPIB / 488` became `GPIB / 488`.
- Error 747 changed from 'Cal checksum failed, HP-IB address' to 'Cal checksum failed, GPIB address'.
- The vector tables moved: bank 0 from $ECBA to $ECC2, bank 1 from $D07C to $D05C, bank 2 from $8410 to $840A and bank 3 from $B4B0 to $B4B8.
- Revision 5.0 has $FF filler at $0100-$061F of every bank where the later images carry a block of $00 and $01 bytes.

The command tree is identical: the same 397 command paths, the same 182 keywords, the same action numbers, the same parameter descriptors. The error table is identical apart from error 747. A program written for revision 5.0 will run unchanged on revision 10.0.

Table 3-8. Summary of Revision Differences

Item	5.0	9.0	10.0
Command tree	identical	identical	identical
Error table	747 says HP-IB	GPIB	GPIB
Front-panel text	HPIB ADDR	GPIB ADDR	GPIB ADDR
Bank 0 content	differs	as 10.0 except revision byte	reference
Bank 1 content	differs	relinked	reference
Bank 2 content	differs	as 10.0 except revision byte	reference
Bank 3 content	differs	as 10.0 except revision byte	reference

4

Instrument Hardware

Signal Flow

The Block Diagram sheet divides the instrument into ten numbered blocks and draws the isolation barrier between block 7, the floating logic, and block 8, the earth-referenced logic. Signal flow from the input terminals is: input and protection, function switching, DC amplifier, A/D converter, main controller. Control flows the other way as serial data: the main controller loads a 72-bit configuration shift-register chain which sets every relay and analogue switch in the measurement path.

There are three serial links on the floating side and one across the barrier:

`Configuration`	Main controller to the 72-bit shift-register chain ending in U601-U606, through the gate array as SERCK, SERDAT and SERSTB. A read-back path SERRBK returns the chain contents; self-test 608, 'Serial configuration readback failed', checks it.
`Front panel`	Main controller to U602, through the gate array as FPDO, FPDI and XFPSK, with FPINT and FPRST direct from U500's port 1.
`Calibration`	Main controller to U505, from the HSO unit as CALCS, CALSK and CALDI, returning on CALDO.
`Isolated`	Main controller to I/O processor, U500's on-chip UART through optocouplers U506 and U704 to U700. Errors 501 and 502 report framing and overrun faults on this link.

Input and Function Switching

Two input channels each present HI and LO terminals. A relay tree selects the channel and the function; a 9.9 M / 100 k divider is switched in for the high-voltage ranges; the ohms current source is routed to the selected terminals for resistance measurement. All of this is driven from the configuration shift register, which is why a fault in the shift register chain presents as a measurement error rather than as a digital fault.

The self-tests that exercise this switching, 613 to 622, are described part by part in the Service Guide, pages 119 to 121.

DC Amplifier

The DC amplifier block carries the input filter, the offset adjustment and the range-select gain switching, and delivers ADIN to the A/D converter. Its three gains and the precharge amplifier, the circuit that prevents the input capacitance from disturbing a high-impedance source, are each separately calibrated and separately self-tested; the Service Guide, pages 120 and 121, gives the procedure for each.

Ohms Current Sources

Five current sources and three compliance clamps are fitted. Their values, and which range uses which, are in the User's Guide, pages 89 and 91; the self-tests that exercise them are in the Service Guide, pages 120 and 121. What is not published is how the firmware reaches them.

The remote interface exposes the clamp as `SENSe:RESistance:VOLTage:LIMit:STATe` and its value as `...:LIMit:VALue`, and the power limit as `SENSe:RESistance:POWer:LIMit:STATe` and `...:VALue`. Both limit branches share the same handler numbers, $83/$84 for the state and $76/$77 for the value. The User's Guide, pages 90 and 91, states that setting either mode cancels the other; the shared handler numbers are the reason it must, and are why errors 1000 and 1001 exist to say so.

Analogue-to-Digital Converter

The converter is a multi-slope integrating type. Its references are derived from a +7 V reference (U403) buffered by U400B and scaled by U400A and U401A into +10REF and -10REF, with REF+0.2 and REF-0.2 taps taken from a precision network inside U102-E. `ADIN` enters through a 100 k resistor in the same network. The switching that steers charge into the integrator is U411, a 74HC4053, driven by `ADSW1` and its complement from the flip-flop U404A, which is clocked by `ADSW` from the gate array.

The main processor participates directly. `SYNC` on P2.7 synchronises the conversion, `COMP` returns the comparator state to the gate array, and the 80C196's own 10-bit converter digitises the residue at the end of the rundown; the Service Guide describes several self-tests as comparing an external measurement with 'the U500 on-chip ADC'. Self-tests 603 to 607 cover the sync pulse, slope convergence, rundown gain and rundown noise.

Integration time is expressed to the user in power-line cycles, so the instrument must know the line frequency, which it measures on `LSENSE` at HSI.0 rather than assuming it. The measured value is readable with `SYSTem:LFRequency?` and, undocumented, with `CALibration:LFRequency?`; both are served by action number $66.

Configuration Data Shift Registers

U601 to U606 are six 74HC4094 eight-bit shift-and-store registers cascaded into a 48-bit section of the chain, producing outputs REG1 to REG48. The chain is clocked by `SERCK`, fed by `SERDATO` and transferred to the output latches by `SERSTB`. The last stage's serial output, `SERDAT1`, returns to the gate array as `SERRBK` so that the whole chain can be read back and compared.

Part of the chain is on the earth-referenced side of the barrier and drives the analogue (chart recorder) output DAC through `AOUT_CLK`, `AOUT_DAT` and `AOUT_STB`, and the injected-current compensation circuit.

The 48 bits above are the six 74HC4094 devices the component locator shows, and they are not the whole chain. The Service Guide's description of self-test 608, page 120, says the test re-sends nine bytes and names the serial path as U501, U101, U150 and U601 through U606. Two further shift-register stages therefore sit on the input and function-switching assemblies, ahead of U601. Read `REG1` to `REG48` in this manual as the six locator-visible stages only; the full configuration word is 72 bits.

Earth-Referenced Logic

U700 is an Intel 87C51FA in a 44-pin carrier, clocked at 12 MHz from the ceramic resonator Y701. It owns the GPIB interface through U701, a TMS9914A with 75ALS160 and 75ALS162 bus transceivers, and the RS-232 interface through U706, an LT1180 transceiver presenting TXD, RXD, DTR and DSR on J704. It also owns the external trigger input J702, the voltmeter-complete output J703, the beeper E700 and the chart-recorder output.

Its only connection to the measurement side is the optically isolated asynchronous serial link, which self-tests 625 and 626 exercise; the Service Guide describes both on page 121. That link is also where the second field of the `*IDN?` revision string comes from, as Chapter 1 shows.

Power Supplies

A single transformer feeds both sides of the barrier. The rails, their tolerances and their test points are tabulated in the Service Guide, page 118, Table 4-1, and are not repeated here. The front panel is the exception that needs explaining: it is supplied with 6 V ac and generates its own rails, and its logic runs between -13 V and -18 V so that the display drivers can swing the vacuum-fluorescent grids without level shifting. That is why the front-panel processor sits outside the isolation barrier and yet cannot share a supply with the rest of the floating section.

Warning



The floating measurement section, including U500, U501, the EPROM, the RAM and the calibration EEPROM, may sit at up to the rated input-to-earth potential of the instrument. Nothing on that side of the barrier may be probed with earth-referenced test equipment while an input is connected.

Integration Apertures

Integration time is set as a number of power-line cycles with `SENSe:VOLTage:DC:NPLCycles` and read back in seconds with `SENSe:VOLTage:DC:APERture?`. The firmware does not compute the aperture: it holds sixteen IEEE-754 single-precision values in bank 0 at $F3DC, eight settings each stored twice, once for a 60 Hz supply and once for 50 Hz, and selects the column from the line frequency it measured at power-on.

Table 4-1. Integration Aperture Table, Bank 0 $F3DC

NPLC	60 Hz	50 Hz	Confirmed
0.02	0.0004 s	0.0004 s	+4.00000000E-04
0.2	0.003 s	0.003 s	+3.00000000E-03
1	0.0166667 s	0.02 s	+2.00000000E-02
2	0.0333333 s	0.04 s	+4.00000000E-02
10	0.166667 s	0.2 s	+2.00000000E-01
20	0.333333 s	0.4 s	+4.00000000E-01
100	1.66667 s	2 s	+2.00000000E+00
200	3.33333 s	4 s	+4.00000000E+00

The two fastest apertures are fixed at 400 µs and 3 ms and do not track the line frequency; the remaining six are exactly N divided by the line frequency. Confirmed on the instrument: it reported `SYSTem:LFRequency? +50`, and stepping `NPLCycles` through all eight settings returned the 50 Hz column of this table value for value.

Note



The User's Guide prints its own version of these figures on page 285, and two of its rows disagree with both the ROM and the instrument. It gives 0.02 NPLC as 334 ms (400 ms), which is a repetition of its own 20 NPLC row, where the instrument returns 400 µs; and it gives 0.2 NPLC as 3 ms (4 ms), where both columns of the ROM hold 3 ms and the instrument returns 3 ms on a 50 Hz supply. Use the table above.

A value of `NPLCycles` that is not one of the eight is not rejected. Confirmed on the instrument: `SENSe:VOLTage:DC:NPLCycles 99` was accepted with no entry in the error queue; the instrument clamps to the nearest supported setting.

Transducer Linearisation Tables

Temperature measurement is done from lookup tables held in bank 2, not from polynomials. Roughly 7 KB is given over to them, reached through a directory of twenty-four words at $A56A: a leading count word, then eight groups of three pointers, one group per thermocouple type, in the order given by the type string `BEJKNRST` at $B5FF.

Each group is three blocks. The *search* block is seventeen six-byte records, each a 32-bit EMF in nanovolts and a 16-bit index, used to narrow the search before interpolation. The *temperature* block is an array of 32-bit temperatures in millidegrees Celsius. The *cold-junction* block is an array of 32-bit EMFs in nanovolts at 1 °C steps beginning at -1 °C.

Table 4-2. Thermocouple Tables, Bank 2

Type	Search	Temp	Cold jn	Points	Range, °C	EMF span, mV
B	$8A48	$8AAE	$8C62	109	249.9 to 1820.0	0.291 to 13.820
E	$8D46	$8DAC	$8EEC	80	-200.0 to 1000.0	-8.825 to 76.373
J	$8FD0	$9036	$93C2	227	-210.0 to 1200.0	-8.095 to 69.553
K	$94A6	$950C	$9740	141	-199.9 to 1372.0	-5.891 to 54.886
N	$9824	$988A	$9A1A	100	-199.9 to 1300.0	-3.990 to 47.513
R	$9AFE	$9B64	$9E50	187	-49.9 to 1768.1	-0.226 to 21.103
S	$9F34	$9F9A	$A22E	165	-49.8 to 1768.0	-0.235 to 18.693
T	$A312	$A378	$A488	68	-200.0 to 400.0	-5.603 to 20.872

The range endpoints reproduce the ranges published in the User's Guide, pages 301 to 303, exactly, and that is what establishes the units. Type K reads 0 nV at 0 °C, 39 474 nV at +1 °C and 78 997 nV at +2 °C against published ITS-90 values of 0, 39.45 µV and 79.00 µV.

Table 4-3. Resistance Thermometer Tables, Bank 2

Alpha	Search	Temp	Points	Range, °C	Resistance span, ohm
0.00385	$B066	$B0CC	76	-200.1 to 849.3	18.490 to 390.284
0.00391	$B1FC	$B262	75	-200.0 to 850.1	17.280 to 394.972

The RTD directory is four words at $B38E and the search key is resistance in units of 10 µohm. The two curves are the alpha of 0.00385 and the alpha of 0.00391 that the User's Guide describes on page 93.

Note



The 34420-90091 Manual Update, page 6, prints the second alpha as `ALPHA = .003916` in its menu diagram, where every edition of the User's Guide gives `.00391`. The ROM settles it: the front-panel string at bank 1 $F7C3 reads `ALPHA=.00391`, and $F7B6 reads `ALPHA=.00385`. Those are the two strings the instrument can display, so `.003916` is an error in the update sheet.

A third RTD type exists and is not a tabulated curve. The 34420-90091 Manual Update adds `USER SPRT`: `SENSe:TEMPerature:TRANsducer:FRTD:TYPE {85|91|USER}` with five ITS-90 deviation-function coefficients set through `...:FRTD:USER:COEFficients`. The firmware carries it: the `TYPE` enumeration in ROM holds three keywords, the `COEFficients` node takes five REAL parameters at actions $93/$94, and the instrument answered `SENS:TEMP:TRAN:FRTD:TYPE? +85` and returned five zeros to `SENS:TEMP:TRAN:FRTD:USER:COEF?`. Because the user curve is computed from coefficients rather than looked up, it has no directory entry, which is why only two curves appear in Table 4-3.

One bench result here goes beyond the published parameter list: `SENSe:TEMPerature:TRANsducer:TYPE` accepts `FRTD` but rejects `RTD` with `-224,"Illegal parameter value"`. Two-wire and three-wire RTDs cannot be selected at all, which matches the instrument's four-wire-only resistance measurement and is not stated in any of the guides.

The directory holds no thermistor curve, but the instrument nonetheless accepts `SENSe:TEMPerature:TRANsducer:TYPE THERmistor` and reads it back, and the front panel offers `THERM 5KO` and a `CH1 THERM` reference junction. The thermistor conversion is therefore computed rather than tabulated. Its coefficients were not located: a search of all three images for Steinhart-Hart constants of the usual 5 k magnitudes, as IEEE singles and doubles in both byte orders, found nothing, so they are presumably immediate operands inside the conversion routine. Chapter 10 lists this as not verified.

5

Remote Interface and Command Parser

Interface Selection

How the two interfaces are selected, and that the choice survives power-off, is in the User's Guide, pages 123 and 124. What that account does not say is where the setting is kept. It is in calibration memory, alongside the GPIB address, and that is why a corrupted address has a calibration checksum error of its own, 747, rather than a configuration error. Chapter 7 gives the region map that follows from the checksum errors.

The parser is a SCPI parser of the period, and it carries two different version numbers. Bank 3 holds the build string `@(#) SCPI Version 2.22` at $F27D, while `SYSTem:VERSion?` returns `1994.0`, the value the User's Guide gives on page 116. The first is the vendor's parser library version and the second the SCPI standard year; only the second is reachable over the bus.

Parser Data Structures

The whole command set is a set of tables in bank 3. There is no per-command code in the parser; a command is a node, a node names a keyword and an action number, and a chain of descriptors says what parameters the action takes. Recovering these tables recovers the command set exactly, including commands the manufacturer did not publish.

Table 5-1. Parser Tables in Bank 3, Revision 10.0

Table	Address	Size	Contents
Keyword text	`$B74C-$BCB5`	182 entries	Short form and long-form remainder, NUL separated
Enumeration lists	`$BCB6-$BDD1`	-	NUL-terminated arrays of keyword pointers
Unit tables	`$BDD2-$BE53`	-	Reversed unit strings with scale codes
Parameter descriptors	`$BE54-$C68F`	20 bytes each	Type, unit table, enumeration list
Command nodes	`$C610-$EDF0`	30 bytes each	Keyword, child list, actions, parameters
Node lists	`$E712-$EA81`	variable	Sibling lists with an implied-node index
Root list	`$EA40`	30 entries	The top level of the tree

The Keyword Table

Keywords are stored as two NUL-terminated strings: the short form, then the characters that extend it to the long form. `SOUR\0CE\0` is SOURce; `WAI\0\0` is WAI, which has no long form. The parser accepts either the short form or the full long form and nothing between, which is the SCPI rule.

The table holds 182 keywords in revision 10.0 and the same 182 in revisions 9.0 and 5.0. Chapter 12 lists them all. They serve both as command mnemonics and as discrete parameter values, which is why `MINimum` and `AUTO` appear in the same table as `CALibration` and `TRIGger`.

Command Nodes

A node is thirty bytes. Where a field is zero the corresponding form of the command does not exist: a node with no command action number rejects the command form with error -113, and a node with no query action number rejects the query form the same way.

Table 5-2. Command Node Layout

Offset	Width	Field
$00	word	Pointer to the keyword
$02	word	Pointer to the child node list, or zero for a leaf
$04	word	Action number for the command form, or zero
$06	word	Head of the command parameter chain, or zero
$08	word	Command flags
$0A	word	Action number for the query form, or zero
$0C	word	Head of the query parameter chain, or zero
$0E	word	Query flags
$10	word	Pointer to the response descriptor, or zero
$12	word	Response item count
$14	word	Index of the implied child, or $FFFF
$16-$1D	8 bytes	Zero in every node examined

Sibling nodes are gathered into a list: a word of list identity, a count, the index of the implied child (or $FFFF if there is none), and then that many node pointers in alphabetical order of long form. The root list is at $EA40 and holds 30 entries:

```
ABORT, CAL, CALCULATE, CALIBRATION, CONFIGURE, DATA, DIAGNOSTIC,
DISPLAY, FETCH, INITIATE, INPUT, L1X, L2X, MEASURE, OUTPUT, READ, ROUTE,
SAMPLE, SENS, SENS1, SENS2, SENSE, SENSE1, SENSE2, STATUS, SYSTEM, TEST,
TRACE, TRIGGER, UNIT
```

Parameter Descriptors

A parameter chain is a linked list; each cell is two words, the address of the next cell and the address of a descriptor. The head of the chain is the first parameter. Descriptors are twenty bytes and live in a single array beginning at $BE54, so a descriptor is shared by every command that takes a parameter of the same kind.

Table 5-3. Parameter Type Codes

Code	Type	Accepted form
`$0002`	Discrete	A keyword from the attached enumeration list, or a boolean
`$0004`	Integer	A decimal integer
`$0006`	Real	A decimal real, optionally with a unit suffix
`$000A`	Real (calibration)	As real, but only accepted when the instrument is unsecured
`$000E`	String	A quoted string
`$0010`	Block	An IEEE 488.2 arbitrary block

Word 2 of a descriptor points to a unit table and word 3 to an enumeration list; either may be zero. The keywords `MINimum`, `MAXimum`, `DEFault`, `AUTO`, `ONCE` and `INFinity` reach a command through its enumeration list like any other discrete value, which is why they are accepted on some parameters and not others and why the command table in Chapter 11 can state exactly which.

Implied Nodes

The third word of a node list is the index within that list of the child that may be omitted. In the root list, whose 30 entries are in alphabetical order, it is 21, the index of `SENSE`, which is why `VOLTage:DC:NPLCycles 10` is accepted without the leading `SENSe:`. The same mechanism supplies `SCALar` under `MEASure` and `CONFigure`, `IMMediate` under `INITiate`, `SEQuence` under `TRIGger` and `WINDow` under `DISPlay`.

Note



A node reached only as an implied node is invisible to a probe that walks the tree by name. The command table in Chapter 11 gives the full path; the abbreviated forms in the User's Guide are what the implied nodes make possible.

A worked case: the serial port

The guides give the RS-232 settings as `SYSTem:COMMunicate:SERial:BAUD` and `SYSTem:COMMunicate:SERial:PARity`. Neither spelling is a path in the recovered tree. What the tree holds is `SERial:RECeive:BAUD`, `SERial:TRANsmit:BAUD`, `SERial:RECeive:PARity:TYPE` and `SERial:TRANsmit:PARity:TYPE`, with `RECeive` as the implied child of `SERial` and `TYPE` as the implied child of `PARity`. The published spelling therefore reaches the receive node, and the transmit spelling is an extra the guides do not mention.

The tree makes it look as though receive and transmit can be set independently. They cannot. Both baud nodes carry actions $37 and $38 and both parity nodes $81 and $82, so the two spellings are one setting. Confirmed on the instrument: all three baud spellings returned `+9600` and all three parity spellings returned `EVEN`.

Unit Suffixes

Unit strings are stored reversed, because the parser matches a suffix from the end of the token: `MHOM` in ROM is MOHM, `LEC` is CEL. The tables recovered are:

Table 5-4. Unit Suffix Tables

Applies to	Accepted suffixes
Voltage	`V`
Resistance	`OHM, MOHM`
Time	`S, SEC`
Current	`A`
Temperature	`C, CEL, F, FAR, K`

The Keithley 181 Emulation Language

How to switch the instrument into its Keithley 181 emulation, and what that emulation is for, are in the User's Guide, pages 221 to 223. The emulation is a complete second parser at $5984 to $5DA6 in bank 2, driven by tables that begin at $B41E. Those tables, which no manufacturer document describes, are the subject of this section.

Table 5-5. Keithley 181 Language Tables, Bank 2

Offset from $B41E	Contents
`+0`	The twelve accepted command letters
`+12`	One argument-state byte per letter
`+24`	A count word
`+26`	Handler address table, twelve words
`+$32`	Numeric character set, `0123456789.-+ E` CR LF
`+$44`	Minimum digit count per letter
`+$50`	Maximum digit count per letter

Table 5-6. Keithley 181 Commands Implemented

Letter	Arg state	Handler	Min digits	Max digits
B	0	$5B57	0	1
D	1	$5B57	0	1
K	1	$5B57	0	1
L	1	$5B4A	1	2
M	1	$5B57	0	1
P	1	$5B57	0	2
R	1	$5B57	1	7
T	1	$5B57	0	3
U	1	$5BAB	0	0
X	0	$5D4E	0	0
Y	0	$5BD4	0	0
Z	2	$5BB4	0	1

Note



The User's Guide, page 223, states that 'all of the commands available for the Keithley 181 are implemented in the 34420A, with the exception of the self-test and calibration commands'. The letter table above says otherwise. Eight further letters are absent, and a letter outside the twelve is rejected by the parser rather than accepted and ignored. A program written for a real 181 will not run unchanged.

The letters not implemented are `F, G, N, O, Q, S, V` and `W`. Most of the twelve that are implemented share the generic handler at $5B57, which converts the ASCII argument to binary and stores it through a pointer table at $B488 into the per-letter variables at $108A to $1095. `L` at $5B4A writes $1095 directly, `U` at $5BAB builds the status string from the pointer list at $B4A0, `Y` at $5BD4 selects a terminator from the table at $B4B2, `Z` at $5BB4 sets the relative-zero flags, and `X` at $5D4E executes. Readings are formatted with `%cDCV%sE%+1d` at $B9DC.

Note



`SYSTem:LANGuage` exists as a command but has no query form: `SYSTem:LANGuage?` returns `-113,"Undefined header"` while `SYSTem:LANGuage` with a bad parameter returns `-102,"Syntax error"`. Both were confirmed on the instrument. The current setting can be read only from the front-panel `I/O MENU` item 5.

6

Undocumented and Service-Only Commands

How These Were Found

The command tree was recovered from the parser tables described in Chapter 5 and compared, path by path, with everything the manufacturer published. That is more than one list, and the difference matters: besides the SCPI command summary on pages 141 to 149 of edition 8 of the User's Guide and its calibration command list, there is a second and substantially different list, the SCPI Compliance Information on pages 224 to 226; there are commands published in earlier editions and dropped from edition 8; and there is the 34420-90091 Manual Update. A command is counted as published here if it appears in any of them. Every path in the recovered tree that carries a command or query action number was then tested on a revision 10.0 instrument by sending the header followed by an invalid parameter and reading the error queue. A header that does not exist gives -113, 'Undefined header'; a header that does exist gives -102, 'Syntax error', and the command is not executed. Each path was tested in its command form, its query form or both, as the tree says it exists (152 test strings in all), and every one of them was present.

The tree holds 397 paths that carry a command or query action number. Six spellings of the `SENSe` root (`SENS`, `SENS1`, `SENS2`, `SENSe`, `SENSe1` and `SENSe2`) carry 47 paths each, and the eleven `CALibration` paths appear twice, under `CAL` and under `CALibration`. Collapsing those leaves 151 distinct commands, of which 115 are published somewhere and 36 are not. One IEEE 488.2 common command, `*_TRG`, is likewise present and unlisted. Table 6-1 is the complete set of the unlisted ones.

The User's Guide acknowledges the gap. Page 224 says that 'many of the required SCPI commands are accepted by the meter but are not described in this manual for simplicity or clarity', and that most of them duplicate a documented command. That accounts for much of Table 6-1. It does not account for `DIAGnostic:PEEK?`, `DIAGnostic:POKE`, `TEST:TEMPerature?` or the calibration extensions, which duplicate nothing.

Caution



The test method above is safe because the parameter is rejected before the action runs; `SYSTem:PRESet @@@` was verified not to reset the instrument. Sending the same headers with valid parameters is a different matter. Several of the commands in this chapter start a calibration or write to memory.

Table 6-1. Commands Absent from the User's Guide

Path	C/Q	Act	Parameters
ABORt	C	$2F	-
CALibration:FLASh	CQ	$3F/$40	-
CALibration:LFRequency	Q	$66	-
CALibration:ZERO:AUTO	CQ	$3C/$3D	INT{OFF|ON|ONCE}
DIAGnostic:PEEK	Q	$53	INT, REAL, REAL
DIAGnostic:POKE	C	$54	INT, REAL, REAL
DIAGnostic:TEMPerature:AMBient	Q	$A7	-
DISPlay:WINDow:TEXT:DATA12	CQ	$19/$1A	DISC
OUTPut:REFerence:OFFSet:AUTO	C	$79	DISC{ONCE}
SENSe:FRESistance:APERture	CQ	$31/$32	REAL{MAXIMUM|MINIMUM}[S|SEC]
SENSe:FRESistance:POWer:LIMit:VALue	CQ	$76/$77	REAL{MAXIMUM|MINIMUM}[V]
SENSe:RESistance:APERture	CQ	$31/$32	REAL{MAXIMUM|MINIMUM}[S|SEC]
SENSe:RESistance:POWer:LIMit:VALue	CQ	$76/$77	REAL{MAXIMUM|MINIMUM}[V]
SENSe:TEMPerature:APERture	CQ	$31/$32	REAL{MAXIMUM|MINIMUM}[S|SEC]
SENSe:VOLTage:DC:APERture	CQ	$31/$32	REAL{MAXIMUM|MINIMUM}[S|SEC]
SENSe:ZERO:AUTO	CQ	$35/$36	INT{OFF|ON|ONCE}
STATus:OPERation:NTRansition	CQ	$26/$27	REAL
STATus:OPERation:PTRansition	CQ	$29/$2A	REAL
STATus:QUEStionable:NTRansition	CQ	$26/$27	REAL
STATus:QUEStionable:PTRansition	CQ	$29/$2A	REAL
SYSTem:BEEPer:IMMediate	C	$39	-
SYSTem:BEEPer:STATe	CQ	$3A/$3B	DISC
SYSTem:CLICk:IMMediate	C	$05	-
SYSTem:CLICk:VOLume	CQ	$06/$07	DISC{NORMAL|SOFT}
SYSTem:COMMunicate:SERial:TRANsmit:BAUD	CQ	$37/$38	INT
SYSTem:COMMunicate:SERial:TRANsmit:PARity:TYPE	CQ	$81/$82	DISC{EVEN|NONE|ODD}
SYSTem:LFRequency	Q	$66	-
SYSTem:LOCS	C	$67	-
SYSTem:LOG	Q	$1C	REAL
SYSTem:LOG10	Q	$1B	REAL
SYSTem:MENU:HELP	CQ	$1F/$20	DISC
SYSTem:REMS	C	$88	-
SYSTem:RWLS	C	$8F	-

Path	C/Q	Act	Parameters
TEST:TEMPerature	Q	$2C	INT, REAL
TRACe:FEED	CQ	$55/$56	DISC{RDG_STORE}, DISC{NONE}
TRACe:POINts	Q	$68	DISC{RDG_STORE}

Several entries are routine. ABORt and the STATus:...:PTRansition pair are ordinary SCPI 1994.0 commands that the instrument implements because its parser does. The SERial:TRANsmit pair is a second spelling of a published setting, as Chapter 5 explains. The ones that matter are grouped in the sections that follow.

Memory Access

DIAGnostic:PEEK?

DIAGnostic:PEEK? <type>,<address>,<count>

Reads the main processor's address space and returns the value as an ASCII number. The three parameters are an integer, a real and a real; the third is required by the parser but was found to have no effect on the result.

Table 6-2. DIAGnostic:PEEK? Type Codes

Type	Width	Returned as	Example at $2000, bank 0
0	byte	unsigned decimal	154
1	word, little-endian	unsigned decimal	9626
2	long, little-endian	unsigned decimal	555623834
3	float	decimal fraction	0.000000

Confirmed on the instrument: DIAG:PEEK? 0,8192,1 returned 154 and DIAG:PEEK? 1,8192,1 returned 9626, which are $9A and $259A, the first byte and word of the interrupt vector table. DIAG:PEEK? 0,65534,1 returned 255, the filler at $FFFE.

The bank in force when the query executes is the bank its handler runs in, not the bank the caller was in. On the instrument tested, DIAG:PEEK? 0,13748,1 (address $35B4, the first byte that differs between banks) returned 240, which is the RET found there in banks 0 and 2, and DIAG:PEEK? 0,15,1 read PORT1 back as 8, whose bank bits are zero. The handler therefore runs in bank 0 and PEEK cannot see banks 1, 2 or 3.

RAM is readable, and that is where PEEK is most useful. The crash buffer is at $1488 (5256), the bank-transfer stack at $17F8 (6136), the gate-array registers at $1FF8 (8184). On the instrument tested $1FF8 read back as 160 and $1FFB as 194.

DIAGnostic:POKE

DIAGnostic:POKE <type>,<address>,<value>

The write counterpart. Its parameter chain is the same three descriptors as PEEK (an integer type code, a real address and a real value), and its action number is $54 where PEEK uses $53. It has no query form.

Caution



This command was not exercised on the instrument used for this work, and nothing here about its behaviour is confirmed. It writes to the address space of the processor that owns the calibration EEPROM, the measurement configuration and the reading memory, and it is not gated on the calibration security state. A wrong address will at best crash the instrument and at worst destroy its calibration. There is no undo.

Diagnostic Measurements

DIAGnostic:TEMPerature:AMBient?

DIAGnostic:TEMPerature:AMBient?

Returns the instrument's internal ambient temperature in degrees Celsius, in the usual signed exponential format. Confirmed on the instrument, which returned `+2.73700000E+01`, that is 27.37 °C. This is the same temperature the instrument records in its calibration message; the message read from the instrument was `" 2 FEB 1998 26.7C"`.

The measurement is made through the internal thermistor used as the thermocouple reference junction, so it reads the temperature of the input terminal block rather than of the air.

TEST:TEMPerature?

TEST:TEMPerature? <real>,<integer>

A two-parameter query, action number $2C. Sent without parameters it returns error -109, 'Missing parameter'; sent with a single integer it returns a syntax error. Its parameter chain is a real followed by an integer, and its response descriptor asks for six items. From firmware analysis alone this is a temperature-conversion service: it appears to convert a transducer reading of the given type into a temperature without disturbing the measurement configuration. Its correct argument values were not determined.

SENSe:...:APERture

SENSe:VOLTage:DC:APERture {<seconds>|MIN|MAX}

SENSe:VOLTage:DC:APERture? [{MIN|MAX}]

The four measurement functions each carry an `APERture` node alongside the documented `NPLCycles` node, with actions $31 and $32 against $6D and $6E. It sets the same integration time expressed in seconds instead of power-line cycles. Confirmed on the instrument, which returned `+2.00000000E-01` for `SENS:VOLT:DC:APER?` while `SENS:VOLT:DC:NPLC?` returned `+1.00000000E+01`. Ten cycles of a 50 Hz supply are 200 ms, so the two agree.

Mathematical Services

Two logarithm queries are present in the tree, take a single real argument and return a real. They are not measurement commands; they compute on the value given. Both were confirmed on the instrument.

Table 6-3. Mathematical Queries

Command	Action	Argument sent	Response
SYSTem:LOG? 100	$1C	100	+4.60517000E+00
SYSTem:LOG10? 100	$1B	100	+2.00000000E+00

4.60517 is the natural logarithm of 100, so LOG is base e and LOG10 base ten. From firmware analysis alone, they appear to be a by-product of the thermistor and thermocouple conversion code, exposed to the parser because the routines were already present.

Sound, Language and Interface Extras

Table 6-4. Unlisted SYSTem Commands

Command	Action	Notes
SYSTem:BEEPer[:IMMediate]	$39	Sounds the beeper once; no parameter
SYSTem:BEEPer:STATe {OFF|ON}	$3A/$3B	Enables the beeper. Confirmed on the instrument, which returned 1
SYSTem:CLICk[:IMMediate]	$05	Sounds the key click once
SYSTem:CLICk:VOLume	$06/$07	Key-click volume
SYSTem:LANGuage	$65	Selects the remote language; command form only. The published route is L1X and L2X, which are documented
SYSTem:LFRequency?	$66	Sensed line frequency. Confirmed; returned +50
SYSTem:MENU:HELP {OFF|ON}	$1F/$20	Front-panel menu help text. Confirmed; returned 1
SYSTem:LOCS / :REMS / :RWLS	$67/$88/$8F	Aliases of the documented LOCal, REMote and RWLock, sharing their action numbers

---

Calibration Extensions

The User's Guide lists nine calibration commands. The tree holds thirteen. The four it does not list are:

CAL:FLASH — Command and query, actions $3F and $40, no parameters. From firmware analysis alone this relates to the FLASH signal on P0.0 of the main processor, which comes from the function-switching sheet, rather than to programmable memory.

CAL:LFR? — Query only, action $66, the same action as the equally unlisted SYSTem:LFRequency?. Returns the sensed line frequency. Confirmed on the instrument, which returned +50.

CAL:ZERO:AUTO — Command and query, actions $3C and $3D, taking OFF, ON or ONCE. Distinct from the equally unlisted SENSe:ZERO:AUTO, which carries actions $35 and $36.

SENS:CURR:BIAS:ZERO:IMM — Command only, action $3E, no parameters. The bias-current zero that the front panel offers as the undocumented 8: BIAS I item of the CAL menu.

Two documented commands are one operation. CALibration:ICURrent? and the implied form of CALibration:ALL?, which is what the documented bare CALibration? reaches, both carry action number $44.

Caution



Nothing in the CALibration subtree was sent to the instrument used for this work except COUNt?, STRing?, SECure:STATe? and LFRequency?. On an unsecured instrument the remainder act immediately.

Front-Panel Menu Extensions

Bank 1 holds two front-panel menu items that the menu maps in the Service Guide, page 48, and the User's Guide, page 50, do not show. One of them is published elsewhere; the other is not.

B: TEMP MENU item 8, 8: SPRT COEF, with its sub-items EDIT A4 through EDIT C7, is documented in the 34420-90091 Manual Update, pages 6 to 8, along with the remote form SENSe:TEMPerature:TRANsducer:FRTD:USER:COEFficients and the USER keyword of ...:FRTD:TYPE. It is missing only from the guides that predate or omit that update. See Transducer Linearisation Tables in Chapter 4.

G: CAL MENU item 8, 8: BIAS I, is not in any of the manufacturer documents examined. The published CAL menu ends at 7: MESSAGE. It corresponds to SENSe:CURRent:BIAS:ZERO:IMMediate, which is itself published, in the User's Guide's SCPI Compliance Information on page 225, though not in its command summary. The front-panel route is the undocumented part.

Note



The item was recovered as text and as a menu record from bank 1 and was not exercised. It sits inside the CAL menu, so reaching it means unsecuring the instrument.

7

Calibration, Security and Non-Volatile Data

Calibration Memory Organisation

Everything the instrument remembers across a power cycle lives in U505, the NMC93C66 serial EEPROM: 256 sixteen-bit words, 512 bytes. The firmware protects it with checksums, and the granularity of those checksums is visible in the error table, because each has its own error number.

There are nine of them, errors 740 to 748, and the message text of each names the region it guards. The messages are printed in the Service Guide, pages 105 and 106, and in the User's Guide, page 241, and are not repeated here; what follows is the inference drawn from them.

Nine regions are checked independently. A failure in one does not prevent the instrument from starting; it raises the corresponding error and the affected function reverts to its default. This is why an instrument with error 747 still measures but forgets its GPIB address, and why error 741 leaves an instrument that will not accept a security code.

The default contents used when a region is initialised are held in bank 2 of the ROM, in a block beginning at $89DF in revision 10.0. The security code is the plainly visible part of it.

Note



The exact word-by-word layout of the EEPROM was not recovered. The nine regions are named by the firmware itself, in the strings that report their failure; the boundaries between them are not.

The Security Code

Calibration is protected by a code held in the EEPROM. The three commands that manage it, their character limits and the procedure for using them are in the Service Guide, pages 68 and 89, and in the User's Guide, page 197. In summary:

CALibration:SECure:STATe {OFF|ON|0|1},<quoted code>
CALibration:SECure:STATe?

CALibration:SECure:CODE <quoted new code>

`CALibration:SECure:STATe` takes two parameters, a discrete state and a string, which is how the code is supplied to unsecure the instrument. `CALibration:SECure:CODE` takes a string only and has no query form, so the code cannot be read back. The query `CALibration:SECure:STATe?` returns 1 when the instrument is secured; the instrument used for this work returned 1 and was left that way.

The factory default

The manufacturer's own statement of the factory code is not consistent across editions, and the ROM settles it. The string `HP034420` appears in bank 2 of all three images, inside the block of EEPROM default values, at these addresses:

Table 7-1. Location of the Default Security Code

Revision	Bank	Address	String
5.0	2	`$89E7`	`HP034420`
9.0	2	`$89E1`	`HP034420`
10.0	2	`$89E1`	`HP034420`

The first edition of the User's Manual, page 107, agrees with the ROM exactly. It gives the factory code as `HP034420`, states that 'the first two characters must be "HP" and the remaining characters must be numbers', shows the entry template as `H P _ _ _ _ _ _`, and prints the example `CAL:SEC:STAT OFF, HP034420`. The second edition says the same.

The third and later Keysight editions replaced every occurrence of `HP` in that passage with `KT`, giving `KT034420` and the rule that the first two characters must be `KT`. The Service Guide followed, and is itself inconsistent: page 89 says `KT034420` and page 68 says `KT34420`, which is a digit short.

Note



The security code is a string in ROM, not a brand. It was not changed when the instrument was rebranded, and the later editions applied a global search and replace to a value that the firmware does not follow. An instrument still holding its factory code answers to `HP034420` and not to `KT034420`. A reader working from a current Keysight edition will fail to unsecure it and may conclude the code has been changed.

Only the last six characters reach the front panel, which is why both spellings appear to work there: the Service Guide's procedure on page 68 shows the display reading `^034420: CODE` and says to enter the six digits alone. The difference matters only on the remote interface, where the whole string is compared.

The firmware evidence is the ROM string and its position in the EEPROM default block; the documentary evidence is the first edition of the User's Manual. It was not tested on the instrument, because testing it means unsecuring it. An instrument whose code has been changed since manufacture will not accept the factory value in any case, and a reader who must recover a lost code has the jumper described below.

The Security Jumper

That a jumper exists, where it is and how to use it to recover a forgotten code are in the Service Guide, pages 89 and 90. What that account does not give is the circuit. JM500 grounds a node pulled up by R504, 5.11 k to +5CV, and that node is sensed on an analogue input of port 0 of U500, not on a digital input. The firmware must therefore run a conversion and compare a threshold before it can decide whether the jumper is fitted.

The security check reports the jumper rather than silently ignoring the code: error 701 is 'Cal security disabled by jumper', so an instrument left with the jumper in place announces it in the error queue.

Caution



Fitting JM500 defeats the calibration security entirely. It is inside the instrument and behind the seal on a calibrated unit. Using it invalidates the calibration certificate even if no constant is changed.

Calibration Count and Message

The instrument counts calibrations and stores a free-text message. Both are readable without unsecuring:

CALibration:COUNt?
CALibration:STRing?

CALibration:STRing <quoted message>

The count advances once per calibration point, so a complete calibration adds roughly 24; that figure and the front-panel routes to both items are in the Service Guide, pages 48 and 92. The instrument used for this work returned `+18` and `" 2 FEB 1998 26.7C"`. The trailing figure in that message is an ambient temperature, and `DIAGnostic:TEMPerature:AMBient?` on the same instrument returned `+2.73700000E+01`. The message is operator-written, not generated, so this is a convention the calibrating laboratory followed rather than something the firmware does.

Calibration Command Sequence

The calibration commands recovered from the tree, with their action numbers, are:

Table 7-2. Calibration Subsystem, Complete

Path	C/Q	Cmd	Qry	Parameters
CALibration:ALL	Q	-	$44	-
CALibration:COUNt	Q	-	$41	-
CALibration:FLASh	CQ	$3F	$40	-
CALibration:ICURrent	Q	-	$44	-
CALibration:LFRequency	Q	-	$66	-
CALibration:OUTPut	CQ	$42	$43	DISC{GAIN|ZERO}
CALibration:SECure:CODE	C	$45	-	STR
CALibration:SECure:STATe	CQ	$46	$47	DISC, STR
CALibration:STRing	CQ	$48	$49	DISC
CALibration:VALue	CQ	$4A	$4B	REAL[V|MOHM|OHM]
CALibration:ZERO:AUTO	CQ	$3C	$3D	INT{OFF|ON|ONCE}

The published adjustment procedure uses `CALibration:VALue` to declare the value of the applied standard and `CALibration?` to perform the step; the Service Guide's chapter 3 gives the sequence of signals for zero, gain, injected current, internal temperature reference and chart output. Nothing in the recovered tree contradicts it. The additions are the four commands listed under Calibration Extensions in Chapter 6.

Caution



Every command in Table 7-2 whose name is not `COUNt?`, `STRing?`, `SECure:STATe?` or `LFRequency?` either performs an adjustment or changes the security state. On a secured instrument they return error 702, 'Cal secured', and do nothing. On an unsecured instrument they act immediately and there is no confirmation step.

8

Diagnostics, Self-Test and Error Codes

Self-Test Structure

What the self-tests do, part by part, is in the Service Guide, pages 119 to 121, and is not repeated here. Two things about the structure are not in that account. The first is the complete list of numbers the front panel will accept, recovered as text from bank 1 of the ROM:

`604, 605, 606, 607, 608, 609, 610, 611, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 627, 630`, together with `ALL` and `OFF`.

The numbers are the same as the error numbers a failure reports, so selecting test 613 runs the 5 µA ohms source test and a failure queues error 613. Tests 601, 602 and 612 are absent from the selectable list, and so are 625 and 626. The Service Guide, page 119, names 601, 625 and 626 as the power-on set, which accounts for three of the five: a test that always runs at power-up needs no menu entry. 602 and 612 are unexplained.

Note



The selectable-test list is a front-panel facility. There is no recovered remote command that runs one numbered test; `*TST?` runs the whole sequence, as the User's Guide, page 113, describes.

The second point is that the firmware's own message strings do not exactly match the published tables. The Service Guide, pages 103 to 106, and the User's Guide, pages 231 to 241, print the error lists in full and should be used as the reference for what each error means. What follows is only the difference between those lists and what is in the ROM.

The Instrument Error Table

There are two error tables, in different banks, and they are structurally identical: a flat array of two-word records, a signed number and a pointer to the message. The one in bank 2 holds the instrument's own 82 codes and is the subject of this section. The one in bank 3 holds the standard SCPI and IEEE 488.2 codes and is the subject of the next.

The bank 2 table is identical in all three firmware revisions apart from the wording of error 747. Its nine negative entries are the standard SCPI errors the measurement code raises directly; every other standard code the instrument can return comes from the parser and has its text in the second table.

The messages themselves are published, in the Service Guide, pages 103 to 106, and in the User's Guide, pages 231 to 241, the latter with a description of each. Rather than reprint them, this section gives what comparison against those lists shows.

Table 8-1. Instrument Errors Not in Any Manufacturer Document

Number	Message in ROM	Remarks
-231	Data questionable	Raised alongside the documented -230. Both set the questionable data summary bit.
724	Cal Low Power FRES offset out of range	Completes the set of offset-calibration errors, 720 to 725, of which the guides print only 720, 722 and 723.
725	Cal Low Voltage FRES offset out of range	As 724, for the voltage-limited ohms mode.

That only three of the 82 entries are unpublished shows that the published error lists are accurate and very nearly complete. The two calibration entries matter because 720 to 725 are consecutive and the guides print an incomplete run, which invites the reader to assume the gaps are unused.

Where the ROM and the printed tables disagree on wording

Table 8-2. Wording Differences

Number	Message in ROM	Message as printed
-230	Data stale	Data corrupt or stale
552	Command not allowed in present function	Command not allowed in current function
747	Cal checksum failed, GPIB address	Cal checksum failed, HP-IB address in revision 5.0

A program that matches on the message text rather than the number will fail on the first two of these. Match on the number.

Errors the Service Guide omits

Five entries at the end of the bank 2 table, 750, 810, 820, 1000 and 1001, are absent from the Service Guide's lists though present in the User's Guide, page 237. They concern the two-channel and temperature facilities and the interaction between the two limited ohms modes. A reader working from the Service Guide alone will not find them.

Errors 1000 and 1001 are informational rather than faults: the instrument accepts the command, turns off the conflicting mode and says so. They are the visible consequence of the two limit branches of the `SENSe:RESistance` subtree sharing handler numbers, as Chapter 4 describes. Error 820 is the only one of the five that the User's Guide lists without a description; from the tables of Chapter 4 it is raised when a reference-junction or user value falls outside the conversion range of the selected transducer, and those ranges are the block endpoints of Tables 4-2 and 4-3.

The Standard SCPI Error Table

A second and entirely separate table in bank 3 at $F0D6 holds the complete set of 100 standard SCPI and IEEE 488.2 codes, from 0 through -440. Its records have the same shape as those of the instrument table, a signed number and a pointer to the message, and it is terminated by a record of $FFFF and $0000. The message strings occupy $F328 to $FAC1. The table is present in all three firmware revisions; in revision 5.0 it begins at $F0CE.

This table is the parser's own vocabulary. An instrument that answers `-113,"Undefined header"` is reading it, not the bank-2 table, and the distinction matters when tracing a fault: an error from this table means the command never reached the measurement code.

It also means the instrument can return errors that no manufacturer document lists, and a program written against the published list will meet them. `-110,"Command header error"` and `-120,"Numeric data error"` are the generic forms the parser falls back on, and neither is printed anywhere.

Most of it has never been printed. Of the 100 entries, 36 appear in one or another of the manufacturer documents and 64 appear in none of them. The tables below mark the published ones with a dot in the `P` column, so that what is new here can be seen at a glance. Six further codes, -160, -161, -168, -170, -171 and -178, are covered in the User's Guide only by the range headings '-160 to -168 Block data errors' and '-170 to -178 Expression errors'; their individual message strings appear nowhere and they are counted here as unpublished.

Table 8-3. Standard Command Errors, 0 and -100 Series

Number	Message	P	Number	Message	P
0	No error	•	-131	Invalid suffix	•
-100	Command error		-134	Suffix too long	•
-101	Invalid character	•	-138	Suffix not allowed	•
-102	Syntax error	•	-140	Character data error	
-103	Invalid separator	•	-141	Invalid character data	•
-104	Data type error	•	-144	Character data too long	•
-105	GET not allowed	•	-148	Character data not allowed	•
-108	Parameter not allowed	•	-150	String data error	
-109	Missing parameter	•	-151	Invalid string data	•
-110	Command header error		-158	String data not allowed	•
-111	Header separator error		-160	Block data error	
-112	Program mnemonic too long	•	-161	Invalid block data	
-113	Undefined header	•	-168	Block data not allowed	
-114	Header suffix out of range		-170	Expression error	
-120	Numeric data error		-171	Invalid expression	
-121	Invalid character in number	•	-178	Expression data not allowed	
-123	Exponent too large	•	-180	Macro error	
-124	Too many digits	•	-181	Invalid outside macro definition	
-128	Numeric data not allowed	•	-183	Invalid inside macro definition	
-130	Suffix error		-184	Macro parameter error	

Table 8-4. Standard Execution Errors, -200 Series

Number	Message	P	Number	Message	P
-200	Execution error		-255	Directory full	
-201	Invalid while in local		-256	File name not found	
-202	Settings lost due to rtl		-257	File name error	
-210	Trigger error		-258	Media protected	
-211	Trigger ignored	•	-260	Expression error	
-212	Arm ignored		-261	Math error in expression	
-213	Init ignored	•	-270	Macro error	
-214	Trigger deadlock	•	-271	Macro syntax error	
-215	Arm deadlock		-272	Macro execution error	
-220	Parameter error		-273	Illegal macro label	
-221	Settings conflict	•	-274	Macro parameter error	
-222	Data out of range	•	-275	Macro definition too long	
-223	Too much data	•	-276	Macro recursion error	
-224	Illegal parameter value	•	-277	Macro redefinition not allowed	
-230	Data corrupt or stale	•	-278	Macro header not found	
-231	Data questionable		-280	Program error	
-240	Hardware error		-281	Cannot create program	
-241	Hardware missing		-282	Illegal program name	
-250	Mass storage error		-283	Illegal variable name	
-251	Missing mass storage		-284	Program currently running	
-252	Missing media		-285	Program syntax error	
-253	Corrupt media		-286	Program runtime error	
-254	Media full				

Table 8-5. Standard Device, System and Query Errors, -300 and -400 Series

Number	Message	P	Number	Message	P
-300	Device specific error		-330	Self-test failed	•
-310	System error		-350	Queue overflow	•
-311	Memory error		-400	Query error	
-312	PUD memory error		-410	Query INTERRUPTED	•
-313	Calibration memory lost		-420	Query UNTERMINATED	•
-314	Save/recall memory lost		-430	Query DEADLOCKED	•
-315	Configuration memory lost		-440	Query UNTERMINATED after indefinite response	•
-321	Out of memory				

Eleven of these were provoked deliberately on the instrument and every returned string matched the extracted table character for character. The stimuli are given in Chapter 10.

Calibration-Failure Front-Panel Captions

When a calibration step fails, the front panel shows `CAL ERROR` for 1.5 seconds and then a twelve-character caption naming the step. The captions are seventeen sixteen-byte records at bank 0 $EF62, each a 16-bit code followed by the caption text, scanned by the display routine at $40E8. Code $49 is handled ahead of the table and displays `CAL ABORTED`.

Note



The 16-bit key is the calibration routine's own return value, not an SCPI error number. No table mapping one to the other exists in the firmware, so a caption must not be read as identifying a particular 7xx error. Read `SYSTem:ERRor?` for that.

Table 8-6. Calibration-Failure Captions, Bank 0 $EF62

Code	Caption	Code	Caption
$4A	CAL VALUE	$4C	BAD FUNC
$4B	CAL SIGNAL	$2E	A/D RUNDOWN
$53	PRECH DAC	$2F	A/D GAIN
$4E	DCV OFFSET	$55	BIAS CURR
$4F	OHM OFFSET	$56	CHARGE COMP
$50	OHM4 OFFSET	$4D	GAIN VALUE
$51	OHM4 OFFSET	$57	INJ CURRENT
$52	OHM4 OFFSET	$46	IS SECURED
$54	-FULL SCALE		

The three records that read `OHM4 OFFSET` are distinct steps with distinct codes; the display has no room to tell them apart. The calibration routine's three callers, at $4177, $41ED and $4204, each test the result of a step and pass it straight to the display routine.

9

Front-Panel Subsystem

Display and Keyboard Hardware

The front-panel assembly 34420-66502 is a self-contained microcomputer. U602 is an Intel 87C51 in a 44-pin carrier running from a ceramic resonator Y601, and it is powered between -13 V and -18 V: VCC on pin 44 is -13 V and ground on pin 22 is -18 V. U601, an MC34064 undervoltage sensor, holds it in reset until the rail is good.

The display U606 is a vacuum-fluorescent panel, part 34420-89301, driven by U603, a ULS75518 serial-input display driver taking `DISP_SDATA`, `DISP_SCLK`, `DISP_LATCH` and `DISP_BLANK` from the processor and producing eighteen anode lines and the grid drives. The keyboard is a matrix scanned with four drive lines on P0.0-P0.3 and four sense lines on P1.0-P1.3, with eight further lines on port 2.

The Front-Panel Link

The main controller talks to U602 through the gate array. On the measurement side the signals are `FPDO` and `FPDI` for data, `XFPSK` for the shift clock, `FPINT` from P1.0 of U500 to request attention and `FPRST` from P1.1 to reset the panel. The gate array raises `CHINT` when the link needs service, which reaches the main processor as INT07.

The handler at $21AC works the link through four gate-array registers in the RAM window:

Table 9-1. Gate-Array Registers Used by the Front-Panel Handler

Address	Access	Use
`$1FF8`	read	Status; bit 4 is polled until the link is ready
`$1FF8`	write	Control; bit 7 is set to start a transfer
`$1FF9`	read/write	Data byte to or from the panel
`$1FFA`	read	Read by the handler at $263E; use not established
`$1FFB`	write	Written with zero to acknowledge the interrupt

The exchange is protected by a timeout scheduled through the high-speed output unit: the handler loads `HSO_COMMAND` with $1A and `HSO_TIME` with the current `TIMER1` plus $48. A panel that stops answering therefore produces a timed event rather than a hang, and the failure surfaces as error 601, 'Front panel does not respond'.

The protocol above the byte level is a small state machine. The handler keeps a state byte in RBAh and a data byte in RBCh, and indexes two tables in the resident region: a four-bit lookup at $219C and a state-transition table at $2188 addressed by `state*2 + data`. Its detailed behaviour was not reconstructed.

Note



U602 is referenced to the negative supply, not to earth or to the input, and the front-panel link does not cross the isolation barrier. The barrier lies between the main controller and the I/O processor only.

Menu Structure

The menu is not a list of strings but a linked graph in bank 1, rooted at $D704. Each record carries a pointer to its parent, to its first child, to its previous and next siblings, and to its thirteen-character display caption; the sibling links are circular, which is why holding a menu key wraps round. Walking the graph from the root reaches 54 records, and those records are the menu.

Table 9-2. Menu Record Fields, Bank 1

Offset	Field
+2	Parent record
+4	First child record
+6	Previous sibling
+8	Next sibling
+10	Pointer to the display caption

The remaining fields of a record hold handler addresses and per-item state and were not decoded. Vertical bars in the stored captions are display-formatting marks, not characters; they are removed below.

Table 9-3. Front-Panel Menu Tree Recovered from ROM

Menu	Record	Item
A: MEAS MENU	$D75A	1:DIG FILTER
	$D7B0	2:INTEGRATE
	$D806	3: OCOMP O
	$D85C	4:LOW POWER O
	$D8B2	5: LOW VOLT O
	$D908	6:LoV LIMIT O
B: TEMP MENU	$D96C	1:PROBE TYPE
	$D9C2	2: UNITS
	$DA18	3: RTD TYPE
	$DA6E	4: RTD Ro
	$DAC4	5: T/C TYPE
	$DB1A	6:COLD JUNCT
	$DB70	7:JUNCT TEMP
	$DBC6	8: SPRT COEF
	$DC1C	8: SPRT COEF
C: MATH MENU	$DC80	1: STATS
	$DCD6	2:NULL VALUE
	$DD2C	3:SCALE GAIN
	$DD82	4:SCALE OFST
D: TRIG MENU	$DDE6	1: READ HOLD
	$DE3C	2:TRIG DELAY
	$DE92	3: N SAMPLES
E: SYS MENU	$DEF6	1:RDGS STORE
	$DF4C	2:SAVED RDGS
	$DFA2	3: ERROR
	$DFF8	4: TEST
	$E04E	5: CHART OUT
	$E0A4	6:CHART SPAN
	$E0FA	7:CHART NULL
	$E150	8: DISPLAY
	$E1A6	9: COMMA
	$E1FC	10: PRESET
	$E252	11: REVISION
F: I/O MENU	$E2B6	1: GPIB ADDR
	$E30C	2: INTERFACE
	$E362	3: BAUD RATE

Menu	Record	Item
	$E3B8	4: PARITY
	$E40E	5: LANGUAGE
G: CAL MENU	$E472	1: UNSECURED
	$E4C8	1: SECURED
	$E51E	2: CALIBRATE
	$E574	3:CHART ZERO
	$E5CA	4:CHART GAIN
	$E620	5:INJECTED I
	$E676	6: CAL COUNT
	$E6CC	7: MESSAGE
	$E722	8: BIAS I

The captions are, with two exceptions, the menu map printed in the Service Guide, page 48, and the User's Guide, page 50; the record addresses are not, and they are what this table adds. `8: SPRT COEF` in the TEMP menu is absent from both maps but is documented in the 34420-90091 Manual Update. `8: BIAS I` in the CAL menu is absent from every manufacturer document examined. Both are discussed in Chapter 6.

Two pairs of records carry the same item number and the firmware chooses between them at run time. `CAL MENU` item 1 is `UNSECURED` or `SECURED` according to the calibration security state, and items 1 to 5 are hidden until the instrument is unsecured, which the Service Guide's map does show, in square brackets. The TEMP menu likewise carries two records for item 8.

Menu Text Recovered from ROM

Most of the parameter-level text below the menu items is the front-panel spelling of a setting the User's Guide already describes, and the guides are the place to look for what each setting does. The four strings below are recorded because the ROM says something about them that the guides do not.

Table 9-4. Front-Panel Text Not Accounted For by the Guides

Displayed	What it shows
`USER SPRT`	The third RTD type, added by the 34420-90091 Manual Update and absent from the earlier guides. Its presence in every one of the three ROM images, including revision 5.0, shows the facility predates the update sheet.
`THERM 5KO`	The thermistor transducer, for which no lookup table exists anywhere in the three images. See Chapter 4.
`CH1 THERM`	Channel 1 used as a thermocouple reference junction, which is the only route by which the thermistor conversion is reached in normal use.
`TESTING....`	Shown while the numbered self-test sequence runs. It is the only front-panel indication that a test selected from `4: TEST` is looping.

Residual Text from a Shared Code Base

Bank 1 contains a block of strings that belong to no 34420A function:

```
AMPL LIMIT   OFST LIMIT   FREQ LIMIT   STOP LIMIT   START LIMIT   FSK
LIMIT   FMDEV LIMIT   %DUTY LIMIT
```

These are the limit messages of a function generator: amplitude, offset, frequency, sweep start and stop, FSK hop frequency, FM deviation and duty cycle. The 34420A has no output and no such parameters. Their presence, along with the strings `LOCKED`, `UNLOCKED` and `NOT DONE YET`, indicates that the front-panel and display code is shared with another instrument of the period and that the unused branches were left in the image rather than compiled out.

The strings are inert, since no recovered menu path reaches them. They are recorded because a search of the ROM for a menu string can otherwise land in code that this instrument never executes.

10

Verification Record

Confirmed by Observation

A 34420A was available on the GPIB at address 22 throughout this work, reporting `HEWLETT-PACKARD,34420A,0,10.0-5.0-2.0`. It was found secured, with a calibration count of 18 and the calibration message `" 2 FEB 1998 26.7C"`, and was left in that state. The following were tested against it.

The command tree

The recovered tree holds 397 paths that carry a command or query action number, which collapse to 151 distinct commands once the six spellings of the `SENSe` root and the two of `CALibration` are set aside. Each was probed in the form or forms the tree says it has (152 test strings) by sending the header followed by an invalid parameter and reading the first entry in the error queue. 152 of 152 returned -102, 'Syntax error', and none returned -113, 'Undefined header'. The recovered tree therefore contains no path the instrument does not implement.

The method was validated first: `SENSe:FOO @@@` returned -113 as expected for an absent header, `SENSe:VOLTage:DC:NPLCycles @@@` returned -102 for a present one, and the instrument's integration time, beeper state and terminal selection were re-read afterwards and were unchanged. The probe therefore does not execute the command it names.

Query-only and command-only forms

Where the recovered node has no command action number the header without a question mark returned -113: `DIAGnostic:PEEK` and `SYSTem:VERSion` both did so, while `DIAG:PEEK?` and `SYST:VERS?` both worked. Where the node has no query action number the reverse held.

Individual commands

Table 10-1. Commands Exercised and Responses Obtained

Command	Response
*IDN?	HEWLETT-PACKARD,34420A,0,10.0-5.0-2.0
SYSTem:VERSion?	1994.0
SYSTem:LFRequency?	+50
CALibration:LFRequency?	+50
SYSTem:LOG? 100	+4.60517000E+00
SYSTem:LOG10? 100	+2.00000000E+00
DIAGnostic:TEMPerature:AMBient?	+2.73700000E+01
SYSTem:MENU:HELP?	1
CALibration:COUNt?	+18
CALibration:STRing?	" 2 FEB 1998 26.7C"
CALibration:SECure:STATe?	1
OUTPut:REFerence:GAIN?	+1.00000000E+03
OUTPut:REFerence:SPAN?	+1.00000000E-03
OUTPut:STATe?	0
INPut:FILTer:LPASs:TYPE?	DIG
INPut:FILTer:LPASs:DIGital:RESPonse?	MED
SENSe:TEMPerature:TRANsducer:TYPE?	FRTD
UNIT:TEMPerature?	C
DATA:FEED?	"CALC"
STATus:OPERation:CONDition?	+0

DIAGnostic:PEEK?

The type codes and the address argument were established by reading locations whose contents are known from the ROM image. `PEEK? 0,8192,1` returned 154 and `PEEK? 1,8192,1` returned 9626, matching $9A and $259A at $2000; `PEEK? 2,8192,1` returned 555623834, which is $211E259A, the first four bytes little-endian; `PEEK? 0,65534,1` returned 255. Nine further addresses in RAM were read without error. The third parameter was varied over 0, 1, 2, 3, 4 and 255 at a fixed address and the returned value did not change.

Firmware revision comparison

The command trees, keyword tables, parameter descriptors and error tables of all three ROM images were extracted independently and compared as data. The trees are identical in all three: the same 397 paths, the same action numbers, the same parameters. The error tables differ in one word of one message.

The standard error table

Eleven of the hundred entries in the bank 3 table were provoked deliberately, each after `*CLS` and a drained error queue. Every returned string matched the extracted table character for character. All eleven are among the codes the User's Guide also prints, pages 231 to 241, so this tests the extraction rather than adding new codes; the unpublished majority of the table could not be provoked deliberately.

Table 10-2. Standard Error Codes Provoked on the Instrument

Sent	Response
`FOO`	`-113,"Undefined header"`
`*IDN` (common command without ?)	`-113,"Undefined header"`
`SENSe:VOLTage:DC:NPLCycles`	`-109,"Missing parameter"`
`SENSe:VOLTage:DC:NPLCycles 1,2`	`-108,"Parameter not allowed"`
`SENSe:VOLTage:DC:NPLCycles 1 XYZ`	`-138,"Suffix not allowed"`
`SENSe:VOLTage:DC:NPLCycles 'a'`	`-158,"String data not allowed"`
`SENSe:VOLTage:DC:NPLCycles #12ab`	`-168,"Block data not allowed"`
`SENSe:VOLTage:DC:NPLCycles 1E999`	`-222,"Data out of range"`
`SENSe:TEMPerature:TRANsducer:TCouple:TYPE X`	`-224,"Illegal parameter value"`
`<header> @@@`	`-102,"Syntax error"`
`query abandoned mid-response`	`-410,"Query INTERRUPTED"`
`query sent with a response pending`	`-420,"Query UNTERMINATED"`

The aperture table

The instrument reported `SYSTem:LFRequency? +50`. `SENSe:VOLTage:DC:NPLCycles` was stepped through all eight settings and `SENSe:VOLTage:DC:APERture?` read at each; all eight returned the 50 Hz column of Table 4-1 exactly. `SENSe:VOLTage:APERture?`, using the implied `DC` node, returned the same value as the full path, which confirms the implied-node handling of Chapter 5 on a live instrument. `NPLCycles` was returned to its as-found value of 10.

Header existence and the language command

The `<header> @@@` probe distinguishes a header that exists, which answers -102 because the header parsed and the parameter did not, from one that does not, which answers -113. On that test `SYSTem:LANGuage` exists as a command while `SYSTem:LANGuage?` returns -113, so the remote language can be selected over the bus but not read back. `SENSe:VOLTage:DC:APERture` and its implied-node form both exist; `SENSe:VOLTage:DC:APERture:ENABle` does not.

Transducer types

The published parameter set is `{TCouple|THERmistor|FRTD}`, User's Guide page 144. What that list does not say is that `RTD` is rejected: it returns -224, so two-wire and three-wire resistance thermometers cannot be selected at all. Both the transducer type and the reference-junction type were found at `FRTD` and `INT` and were returned to them.

Not Verified

The following are asserted from firmware analysis alone.

- The behaviour of `DIAGnostic:POKE`. Its parameter types are read from the parser tables; it was not sent to the instrument. Its argument order is assumed to match `PEEK`.
- The behaviour of `CALibration:FLASh`, `CALibration:OUTPut`, `CALibration:ZERO:AUTO`, `CALibration:ICURrent?`, `CALibration:ALL?` and `SENSe:CURRent:BIAS:ZERO:IMMediate`. None was sent, because all of them act on calibration.
- That `HP034420` still unsecures an instrument holding its factory code. The ROM string, the first and second editions of the User's Manual and the Service Guide's own front-panel illustration all agree on it, but entering it means unsecuring the instrument, which was not done.
- The argument values for `TEST:TEMPerature?`. The command exists and takes two parameters; the values it accepts were not found.
- The word-by-word layout of the calibration EEPROM. Only the names and the existence of its nine checksummed regions are established.
- The meaning of the seventy-byte bank header beyond its first two bytes. Those two are the major and minor firmware revision; the remaining sixty-eight are several ten-element per-function boolean arrays whose individual meanings were not established.
- The RAM paging discipline. RAMA13 exists and is driven from P2.6; which page holds which variable was not traced.
- The coefficients of the thermistor conversion. The instrument accepts `THERmistor` as a transducer type but no thermistor curve exists in the transducer directory and no Steinhart-Hart constants were found in any of the three images.
- The value lists a front-panel menu leaf offers. The menu graph is walked and every caption read, but the child pointer of a leaf addresses a structure that is not another menu record and was not decoded.
- The correspondence between the calibration-failure captions of Table 8-6 and the 7xx error numbers. The captions are keyed by the calibration routine's return value and no mapping table exists in the firmware.
- Whether U500 is an 80C196KB or an 80C196KC. The schematic says KB; the firmware uses the window select register at SFR $14, which is a KC feature. See Chapter 2.
- The detailed front-panel link protocol above the byte level.
- The claim that INT13 is EXTINT1 and INT11 timer 2 capture. The transmit and receive assignments of INT08 and INT09 are established from the firmware's own mask manipulation; the remaining upper vectors are named from the 80C196KB data sheet and could be one position out.

Two of these would settle quickly on an instrument that may be unsecured: the security code, by entering 034420 at the front panel, and the EEPROM layout, by reading the device with `DIAG:PEEK?` before and after a calibration step and differencing.

How to Reproduce

The command-tree probe needs only a controller and a VISA library. The method is:

1. Send the header under test followed by a space and three at-signs, which cannot begin any legal parameter.

2. Read `SYSTem:ERRor?` once.
3. Send `*CLS` to clear anything left in the queue.
4. A first error of -113 means the header does not exist. Anything else (in practice -102) means it does.

The ROM tables are reached from three constants. In each bank, the word at $2164 is that bank's inter-bank vector table base; the byte at that address less $46 is the firmware revision; and in bank 3 the SCPI keyword text begins at the first run of upper-case letters above the vector table. From the keyword bounds, every command node is a 30-byte record beginning with a pointer into that range, and the root list is the node list that no node points at.

The extraction scripts used for this manual walk exactly that path and take no address as given except $2164, which is fixed by the resident code. They therefore work unchanged on any of the three images.

11

Command Reference

How to Read an Entry

The command tables in this chapter have five columns:

`Path`	The full path with implied nodes shown. Short forms are in upper case. The `SENSe` root may be omitted, and `SENSe1` and `SENSe2` select the channel; the table lists the family once.
`C/Q`	`C` if the command form exists, `Q` if the query form exists.
`Act`	The action numbers the node carries, command form first. These are the firmware's own dispatch numbers; two commands with the same number are the same operation reached by two names.
`Parameters`	The parameter types in order, from the descriptor chain. Braces list the accepted keywords, square brackets the accepted unit suffixes.
`U`	`U` where the command appears in none of the manufacturer documents: the User's Guide's SCPI command summary, its separate SCPI Compliance Information on pages 224 to 226, the earlier editions of the User's Manual, or the 34420-90091 Manual Update. Blank where it is published in at least one of them.

Action numbers shared between commands are the most informative column in the tables. They are how `CALibration:ICURrent?` was identified as an alias of `CALibration:ALL?`, and how the voltage-limit and power-limit branches of the ohms subtree were shown to be one mechanism.

Complete Command Table

The tables that follow are the complete set of commands the parser accepts, generated from the tables in bank 3 of the revision 10.0 ROM. They are therefore exhaustive: a command not in them does not exist, and every command in them was confirmed present on the instrument. The `SENSe` family is listed once; `SENSe1` and `SENSe2` select the channel and `SENS`, `SENS1` and `SENS2` are accepted spellings of the same three roots.

Table 11-1. IEEE 488.2 Common Commands

Path	C/Q	Act	Parameters	U
*CLS	C	$08	-	
*ESE	CQ	$09/$0A	INT	
*ESR	Q	$0B	-	
*IDN	Q	$0C	-	
*OPC	CQ	$0D/$0E	-	
*PSC	CQ	$0F/$10	INT	
*RST	C	$11	-	
*SRE	CQ	$12/$13	INT	
*STB	Q	$14	-	
*TRG	C	$4C	-	
*TST	Q	$4D	-	
*WAI	C	$15	-	
*_TRG	C	$92	-	U

The common commands live in their own node list at $E712, outside the root list, which is why they are reached with a leading asterisk rather than a colon. *WAI is present but unlisted in the User's Guide, and the thirteenth entry, _TRG, cannot be reached at all: the lexer rejects the leading underscore with -101, 'Invalid character', whether it is written *_TRG or _TRG. It is presumably reachable in the Keithley 181 language. *SAV, *RCL, *OPT and *LRN are not implemented; all four return -113.

Table 11-2. ABORt, READ, FETCh and INITiate Commands

Path	C/Q	Act	Parameters	U
ABORt	C	$2F	-	U
FETCh	Q	$57	-	
INITiate:IMMediate	C	$64	-	
L1X	C	$30	-	
L2X	C	$30	-	
READ	Q	$87	-	
TEST:TEMPerature	Q	$2C	INT, REAL	U

Table 11-3. CALCulate Commands

Path	C/Q	Act	Parameters	U
CALCulate:AVERage:AVERage	Q	$1E	-	
CALCulate:AVERage:COUNt	Q	$1E	-	
CALCulate:AVERage:MAXimum	Q	$1E	-	
CALCulate:AVERage:MINimum	Q	$1E	-	
CALCulate:AVERage:PTPeak	Q	$1E	-	
CALCulate:AVERage:SDEViation	Q	$1E	-	
CALCulate:FUNCtion	CQ	$01/$02	DISC{AVERAGE|SCALE}	
CALCulate:SCALe:GAIN	CQ	$1D/$1E	REAL[V|MOHM|OHM]	
CALCulate:SCALe:OFFSet	CQ	$1D/$1E	REAL[V|MOHM|OHM]	
CALCulate:STATe	CQ	$03/$04	DISC	

Table 11-4. CALibration Commands

Path	C/Q	Act	Parameters	U
CALibration:ALL	Q	$44	-	
CALibration:COUNt	Q	$41	-	
CALibration:FLASh	CQ	$3F/$40	-	U
CALibration:ICURrent	Q	$44	-	
CALibration:LFRequency	Q	$66	-	U
CALibration:OUTPut	CQ	$42/$43	DISC{GAIN|ZERO}	
CALibration:SECure:CODE	C	$45	STR	
CALibration:SECure:STATe	CQ	$46/$47	DISC, STR	
CALibration:STRing	CQ	$48/$49	DISC	
CALibration:VALue	CQ	$4A/$4B	REAL[V|MOHM|OHM]	
CALibration:ZERO:AUTO	CQ	$3C/$3D	INT{OFF|ON|ONCE}	U

Table 11-5. CONFigure and MEASure Commands

Path	C/Q	Act	Parameters	U
CONFigure	Q	$51	-	
CONFigure:SCALar:FRESistance	C	$50	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[MOHM|OHM], REAL{DEFAULT|MAXIMUM|MINIMUM}[MOHM|OHM], BLOCK	
CONFigure:SCALar:RESistance	C	$50	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[MOHM|OHM], REAL{DEFAULT|MAXIMUM|MINIMUM}[MOHM|OHM], BLOCK	
CONFigure:SCALar:TEMPerature	C	$52	DISC{DEFAULT|FRTD|TCOUPLE|THERMISTOR}, INT{B|DEFAULT|E|J|K|N|R|S|T|USER}, REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[C|CEL|F|FAR|K], REAL{DEFAULT|MAXIMUM|MINIMUM}[C|CEL|F|FAR|K]	
CONFigure:SCALar:VOLTage:DC	C	$50	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[V], REAL{DEFAULT|MAXIMUM|MINIMUM}[V], BLOCK	
CONFigure:SCALar:VOLTage:DC:DIFFerence	C	$4E	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[V], REAL{DEFAULT|MAXIMUM|MINIMUM}[V]	
CONFigure:SCALar:VOLTage:DC:RATio	C	$4F	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[V], REAL{DEFAULT|MAXIMUM|MINIMUM}[V]	
MEASure:SCALar:FRESistance	Q	$6B	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[MOHM|OHM], REAL{DEFAULT|MAXIMUM|MINIMUM}[MOHM|OHM], BLOCK	
MEASure:SCALar:RESistance	Q	$6B	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[MOHM|OHM], REAL{DEFAULT|MAXIMUM|MINIMUM}[MOHM|OHM], BLOCK	
MEASure:SCALar:TEMPerature	Q	$6C	DISC{DEFAULT|FRTD|TCOUPLE|THERMISTOR}, INT{B|DEFAULT|E|J|K|N|R|S|T|USER}, REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[C|CEL|F|FAR|K], REAL{DEFAULT|MAXIMUM|MINIMUM}[C|CEL|F|FAR|K]	

Path	C/Q	Act	Parameters	U
MEASure:SCALar:VOLTage:DC	Q	$6B	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[V], REAL{DEFAULT|MAXIMUM|MINIMUM}[V], BLOCK	
MEASure:SCALar:VOLTage:DC:DIFFerence	Q	$69	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[V], REAL{DEFAULT|MAXIMUM|MINIMUM}[V]	
MEASure:SCALar:VOLTage:DC:RATio	Q	$6A	REAL{AUTO|DEFAULT|MAXIMUM|MINIMUM}[V], REAL{DEFAULT|MAXIMUM|MINIMUM}[V]	

Table 11-6. DATA and TRACe Commands

Path	C/Q	Act	Parameters	U
DATA:FEED	CQ	$55/$56	DISC{RDG_STORE}, DISC{NONE}	
DATA:POINts	Q	$68	DISC{RDG_STORE}	
TRACe:FEED	CQ	$55/$56	DISC{RDG_STORE}, DISC{NONE}	U
TRACe:POINts	Q	$68	DISC{RDG_STORE}	U

Table 11-7. DIAGnostic Commands

Path	C/Q	Act	Parameters	U
DIAGnostic:PEEK	Q	$53	INT, REAL, REAL	U
DIAGnostic:POKE	C	$54	INT, REAL, REAL	U
DIAGnostic:TEMPerature:AMBient	Q	$A7	-	U

Table 11-8. DISPlay Commands

Path	C/Q	Act	Parameters	U
DISPlay:WINDow:STATe	CQ	$17/$18	DISC	
DISPlay:WINDow:TEXT:CLEar	C	$16	-	
DISPlay:WINDow:TEXT:DATA	CQ	$19/$1A	DISC	
DISPlay:WINDow:TEXT:DATA12	CQ	$19/$1A	DISC	U

Table 11-9. INPut Commands

Path	C/Q	Act	Parameters	U
INPut:FILTer:LPASs:DIGital:PRECharge:AUTO:STATe	CQ	$5A/$5B	DISC	
INPut:FILTer:LPASs:DIGital:RESPonse	CQ	$58/$59	DISC{FAST|MEDIUM|SLOW}	
INPut:FILTer:LPASs:STATe	CQ	$5C/$5D	DISC	
INPut:FILTer:LPASs:TYPE	CQ	$5E/$5F	DISC{ANALOG|BOTH|DIGITAL}	

Table 11-10. OUTPut Commands

Path	C/Q	Act	Parameters	U
OUTPut:REFerence:GAIN	Q	$78	-	
OUTPut:REFerence:OFFSet	CQ	$7A/$7B	REAL{MAXIMUM|MINIMUM}[V|MOHM|OHM]	
OUTPut:REFerence:OFFSet:AUTO	C	$79	DISC{ONCE}	U
OUTPut:REFerence:OFFSet:NULL	C	$79	DISC{ONCE}	
OUTPut:REFerence:SPAN	CQ	$7C/$7D	REAL{MAXIMUM|MINIMUM}[V|MOHM|OHM]	
OUTPut:REFerence:VALue	C	$80	REAL{MAXIMUM|MINIMUM}[V]	
OUTPut:STATe	CQ	$7E/$7F	DISC	

Table 11-11. ROUTe, SAMPle and TRIGger Commands

Path	C/Q	Act	Parameters	U
ROUTe:TERMinals	CQ	$AA/$AB	DISC{FRON1|FRON2|FRONT|FRONT1|FRONT2}	
SAMPle:COUNt	CQ	$90/$91	REAL{MAXIMUM|MINIMUM}	
TRIGger:SEQuence:COUNt	CQ	$95/$96	REAL{INFINITY|MAXIMUM|MINIMUM}	
TRIGger:SEQuence:DELay	CQ	$99/$9A	REAL{MAXIMUM|MINIMUM}[S|SEC]	
TRIGger:SEQuence:DELay:AUTO	CQ	$97/$98	DISC	
TRIGger:SEQuence:SOURce	CQ	$9B/$9C	DISC{BUS|EXTERNAL|IMMEDIATE}	

Table 11-12. SENSe Commands

Path	C/Q	Act	Parameters	U
SENSe:CURRent:BIAS:ZERO:IMMediate	C	$3E	-	
SENSe:FRESistance:APERture	CQ	$31/$32	REAL{MAXIMUM|MINIMUM}[S|SEC]	U
SENSe:FRESistance:NPLCycles	CQ	$6D/$6E	REAL{MAXIMUM|MINIMUM}	
SENSe:FRESistance:NULL:STATe	CQ	$70/$71	DISC	
SENSe:FRESistance:NULL:VALue	CQ	$72/$73	REAL{MAXIMUM|MINIMUM}[MOHM|OHM]	
SENSe:FRESistance:OCOMpensated	CQ	$74/$75	DISC	
SENSe:FRESistance:POWer:LIMit:STATe	CQ	$83/$84	DISC	
SENSe:FRESistance:POWer:LIMit:VALue	CQ	$76/$77	REAL{MAXIMUM|MINIMUM}[V]	U
SENSe:FRESistance:RANGe:AUTO	CQ	$33/$34	DISC	
SENSe:FRESistance:RANGe:UPPer	CQ	$85/$86	REAL{MAXIMUM|MINIMUM}[MOHM|OHM]	
SENSe:FRESistance:RESolution	CQ	$89/$8A	REAL{MAXIMUM|MINIMUM}[MOHM|OHM]	
SENSe:FRESistance:VOLTage:LIMit:STATe	CQ	$83/$84	DISC	
SENSe:FRESistance:VOLTage:LIMit:VALue	CQ	$76/$77	REAL{MAXIMUM|MINIMUM}[V]	
SENSe:FUNCtion:ON	CQ	$60/$61	DISC	
SENSe:NULL	CQ	$6F/$71	DISC{OFF|ON|ONCE}	
SENSe:NULL:VALue	CQ	$72/$73	REAL{MAXIMUM|MINIMUM}[MOHM|OHM]	
SENSe:RESistance:APERture	CQ	$31/$32	REAL{MAXIMUM|MINIMUM}[S|SEC]	U
SENSe:RESistance:NPLCycles	CQ	$6D/$6E	REAL{MAXIMUM|MINIMUM}	
SENSe:RESistance:NULL:STATe	CQ	$70/$71	DISC	
SENSe:RESistance:NULL:VALue	CQ	$72/$73	REAL{MAXIMUM|MINIMUM}[MOHM|OHM]	
SENSe:RESistance:OCOMpensated	CQ	$74/$75	DISC	
SENSe:RESistance:POWer:LIMit:STATe	CQ	$83/$84	DISC	
SENSe:RESistance:POWer:LIMit:VALue	CQ	$76/$77	REAL{MAXIMUM|MINIMUM}[V]	U
SENSe:RESistance:RANGe:AUTO	CQ	$33/$34	DISC	
SENSe:RESistance:RANGe:UPPer	CQ	$85/$86	REAL{MAXIMUM|MINIMUM}[MOHM|OHM]	
SENSe:RESistance:RESolution	CQ	$89/$8A	REAL{MAXIMUM|MINIMUM}[MOHM|OHM]	
SENSe:RESistance:VOLTage:LIMit:STATe	CQ	$83/$84	DISC	

Path	C/Q	Act	Parameters	U
SENSe:RESistance:VOLTage:LIMit:VALue	CQ	$76/$77	REAL{MAXIMUM|MINIMUM}[V]	
SENSe:TEMPerature:APERture	CQ	$31/$32	REAL{MAXIMUM|MINIMUM}[S|SEC]	U
SENSe:TEMPerature:NPLCycles	CQ	$6D/$6E	REAL{MAXIMUM|MINIMUM}	
SENSe:TEMPerature:NULL:STATe	CQ	$70/$71	DISC	
SENSe:TEMPerature:NULL:VALue	CQ	$A3/$A4	REAL{MAXIMUM|MINIMUM}[C|CEL|F|FAR|K]	
SENSe:TEMPerature:TRANsducer:FRTD:RESistance:REFerence	CQ	$8B/$8C	REAL{MAXIMUM|MINIMUM}[MOHM|OHM]	
SENSe:TEMPerature:TRANsducer:FRTD:TYPE	CQ	$8D/$8E	INT{USER}	
SENSe:TEMPerature:TRANsducer:FRTD:USER:COEFficients	CQ	$93/$94	REAL, REAL, REAL, REAL, REAL	
SENSe:TEMPerature:TRANsducer:TCouple:RJUNction:TYPE	CQ	$9D/$9E	DISC{FIXED|INTERNAL|THERMISTOR}	
SENSe:TEMPerature:TRANsducer:TCouple:RJUNction:VALue	CQ	$9F/$A0	REAL{MAXIMUM|MINIMUM}[C|CEL|F|FAR|K]	
SENSe:TEMPerature:TRANsducer:TCouple:TYPE	CQ	$A1/$A2	DISC{B|DEFAULT|E|J|K|N|R|S|T}	
SENSe:TEMPerature:TRANsducer:TYPE	CQ	$A5/$A6	DISC{DEFAULT|FRTD|TCOUPLE|THERMISTOR}	
SENSe:VOLTage:DC:APERture	CQ	$31/$32	REAL{MAXIMUM|MINIMUM}[S|SEC]	U
SENSe:VOLTage:DC:NPLCycles	CQ	$6D/$6E	REAL{MAXIMUM|MINIMUM}	
SENSe:VOLTage:DC:NULL:STATe	CQ	$70/$71	DISC	
SENSe:VOLTage:DC:NULL:VALue	CQ	$72/$73	REAL{MAXIMUM|MINIMUM}[V]	
SENSe:VOLTage:DC:RANGe:AUTO	CQ	$33/$34	DISC	
SENSe:VOLTage:DC:RANGe:UPPer	CQ	$85/$86	REAL{MAXIMUM|MINIMUM}[V]	
SENSe:VOLTage:DC:RESolution	CQ	$89/$8A	REAL{MAXIMUM|MINIMUM}[V]	
SENSe:ZERO:AUTO	CQ	$35/$36	INT{OFF|ON|ONCE}	U

Table 11-13. STATus Commands

Path	C/Q	Act	Parameters	U
STATus:OPERation:CONDition	Q	$22	-	
STATus:OPERation:ENABle	CQ	$23/$24	REAL	
STATus:OPERation:EVENt	Q	$25	-	
STATus:OPERation:NTRansition	CQ	$26/$27	REAL	U
STATus:OPERation:PTRansition	CQ	$29/$2A	REAL	U
STATus:PRESet	C	$28	-	
STATus:QUEStionable:CONDition	Q	$22	-	
STATus:QUEStionable:ENABle	CQ	$23/$24	REAL	
STATus:QUEStionable:EVENt	Q	$25	-	
STATus:QUEStionable:NTRansition	CQ	$26/$27	REAL	U
STATus:QUEStionable:PTRansition	CQ	$29/$2A	REAL	U

Table 11-14. SYSTem Commands

Path	C/Q	Act	Parameters	U
SYSTem:BEEPer:IMMediate	C	$39	-	U
SYSTem:BEEPer:STATe	CQ	$3A/$3B	DISC	U
SYSTem:CLICk:IMMediate	C	$05	-	U
SYSTem:CLICk:VOLume	CQ	$06/$07	DISC{NORMAL|SOFT}	U
SYSTem:COMMunicate:SERial:RECeive:BAUD	CQ	$37/$38	INT	
SYSTem:COMMunicate:SERial:RECeive:PARity:TYPE	CQ	$81/$82	DISC{EVEN|NONE|ODD}	
SYSTem:COMMunicate:SERial:TRANsmit:BAUD	CQ	$37/$38	INT	U
SYSTem:COMMunicate:SERial:TRANsmit:PARity:TYPE	CQ	$81/$82	DISC{EVEN|NONE|ODD}	U
SYSTem:ERRor	Q	$2B	-	
SYSTem:INTerface:TYPE	CQ	$62/$63	DISC{HPIB|RS232}	
SYSTem:LANGuage	C	$65	DISC	
SYSTem:LFRequency	Q	$66	-	U
SYSTem:LOCal	C	$67	-	
SYSTem:LOCS	C	$67	-	U
SYSTem:LOG	Q	$1C	REAL	U
SYSTem:LOG10	Q	$1B	REAL	U
SYSTem:MENU:HELP	CQ	$1F/$20	DISC	U
SYSTem:PRESet	C	$21	-	
SYSTem:REMote	C	$88	-	
SYSTem:REMS	C	$88	-	U
SYSTem:RWLock	C	$8F	-	
SYSTem:RWLS	C	$8F	-	U
SYSTem:VERSion	Q	$2D	-	

Table 11-15. UNIT Commands

Path	C/Q	Act	Parameters	U
UNIT:TEMPerature	CQ	$A8/$A9	DISC{C|CEL|F|FAR|K}	

Excluding the IEEE 488.2 common commands of Table 11-1, the tables hold 151 commands, of which 115 are published in one or another manufacturer document and 36 are marked U. Restoring the CAL spelling of CALibration and the five alternative spellings of the SENSe root gives the 397 action-carrying paths the tree contains.

Command Entries

An entry is given where the recovered behaviour adds something to what the manufacturer published: a command that appears in none of their documents, or one that does appear but whose action numbers, parameter set or front-panel route say more than the published description. A command the guides cover adequately has no entry here; the complete command table above lists it, and the User's Guide describes it. Entries are in alphabetical order.

ABORt

ABORt

Aborts a measurement in progress and returns the trigger system to the idle state. Action number $2F, no parameters, no query form, and it sits at the root of the tree. Standard SCPI, and the 34401A documents it; the 34420A User's Guide does not list it, but the instrument implements it.

Action $2F • root level • Published in no manufacturer document examined.

CALibration:ALL?

CALibration:ALL?

Performs the complete calibration sequence for the present function and range and returns 0 for success or 1 for failure. Shares action number $44 with `CALibration:ICURrent?`, which is therefore the same operation under a second name.

On a secured instrument the command returns error 702, 'Cal secured', and does nothing. On an unsecured instrument it acts at once and advances the calibration count.

Caution



Not exercised during this work. On an unsecured instrument this command overwrites calibration constants.

Action $44 • requires UNSECURED • Reached by the documented bare `CALibration?` *as the implied child of the CALibration node.*

CALibration:FLASH

CALibration:FLASH

CALibration:FLASH?

Command and query with no parameters, actions $3F and $40. Both forms were confirmed present on the instrument; neither was executed.

The name is shared with the `FLASH` net on the schematic, which runs from the function-switching sheet to P0.0 of U500, an analogue housekeeping input, not a memory-programming control. From firmware analysis alone this command relates to that signal and not to reprogramming the EPROM, which is a socketed one-time device in any case.

Actions $3F / $40 • CALibration subtree • Published in no manufacturer document examined.

CALibration:ICURrent?

CALibration:ICURrent?

Query only, documented in the User's Guide as the injected-current calibration. It carries action number $44, the same number as `CALibration:ALL?` and therefore as the bare `CALibration?`, so the three names reach one routine. The User's Guide describes them as separate operations; the firmware does not distinguish them.

Caution



Because it is an alias of `CALibration:ALL?`, sending it to an unsecured instrument starts a calibration.

Action $44 • requires UNSECURED • Published by the manufacturer; see the text for where.

CALibration:LFRequency?

CALibration:LFRequency?

Returns the line frequency the instrument has sensed on the `LSENSE` input, in hertz. Carries action number $66, the same as the documented `SYSTem:LFRequency?`, and returns the same value.

Confirmed on the instrument: both forms returned `+50`.

Action $66 • no security gate • Published in no manufacturer document examined.

CALibration:OUTPut

CALibration:OUTPut {GAIN|ZERO}

CALibration:OUTPut?

Described in the User's Guide, page 197, and adjusted through the procedure in the Service Guide. Recorded here only for its action numbers, $42 and $43, and because the recovered parameter descriptor shows a discrete accepting exactly `GAIN` and `ZERO` and nothing else, not even `MINimum` or `MAXimum`.

Actions $42 / $43 • requires UNSECURED • Published by the manufacturer; see the text for where.

CALibration:SECure:CODE

CALibration:SECure:CODE <new code>

Replaces the stored security code. The character limits, the front-panel prefix rule and the recovery procedure are in the Service Guide, pages 89 and 90.

The firmware facts are that the node carries action $45, that it takes a single string parameter, and that it has no query form at all: there is no query action number, so the code cannot be read back even on an unsecured instrument. Chapter 7 gives the factory default as it stands in ROM, which is not the value the guides print.

Caution



A code set remotely and then forgotten can only be cleared by fitting JM500 inside the instrument.

Action $45 • requires UNSECURED • Published by the manufacturer; see the text for where.

CALibration:ZERO:AUTO

CALibration:ZERO:AUTO {OFF|ON|ONCE}

CALibration:ZERO:AUTO?

Actions $3C and $3D. The parameter descriptor is an integer with the enumeration `OFF`, `ON`, `ONCE`, the same descriptor the documented `SENSe:ZERO:AUTO` uses, but with a different action number, so it is a distinct control.

From firmware analysis alone this is the autozero state used during calibration rather than during measurement. The query returned a VISA timeout on the secured instrument tested, which is consistent with the action being gated on the security state.

Actions $3C / $3D • CALibration subtree • Published in no manufacturer document examined.

DIAGnostic:PEEK?

DIAGnostic:PEEK? <type>,<address>,<count>

Reads the main processor's address space. `<type>` is 0 for a byte, 1 for a 16-bit word, 2 for a 32-bit long and 3 for a floating-point value; `<address>` is a decimal address in the 64 KB space; `<count>` is required by the parser and was found to have no effect. The value is returned as a decimal number, unsigned for the integer types.

The read is made in whatever ROM bank the handler runs in. On the instrument tested that is bank 0, established by reading $35B4 (the first address whose contents differ between banks) and by reading PORT1 at address 15, which returned 8. Banks 1, 2 and 3 are therefore not reachable with this command.

RAM, the register file and the special-function registers are all readable, which makes this the practical way to inspect the crash buffer at $1488, the bank-transfer stack at $17F8 and the gate-array registers at $1FF8.

Note



Addresses are given in decimal. $2000 is 8192, $1488 is 5256, $17F8 is 6136, $1FF8 is 8184.

Action $53 • no security gate • Published in no manufacturer document examined.

DIAGnostic:POKE

DIAGnostic:POKE <type>,<address>,<value>

Writes to the main processor's address space. The parameter chain is the same three descriptors as `DIAGnostic:PEEK?` (an integer type code, a real address and a real value), and the type codes are assumed to have the same meaning. Action number $54. There is no query form.

This command was not sent to the instrument used for this work. Its existence and its parameter types are established; its behaviour is not.

Caution



This command writes to the address space of the processor that owns the calibration EEPROM, the measurement configuration and the reading memory, and it is not gated on the calibration security state. A wrong address will crash the instrument or destroy its calibration, and there is no confirmation and no undo.

Action $54 • no security gate • Published in no manufacturer document examined.

DIAGnostic:TEMPerature:AMBient?

DIAGnostic:TEMPerature:AMBient?

Returns the instrument's internal temperature in degrees Celsius. Action number $A7. Confirmed on the instrument, which returned `+2.73700000E+01`.

The sensor is the internal thermistor that also serves as the thermocouple reference junction, so the value is the temperature of the input terminal block. The calibration message on the instrument used for this work ended in `26.7C`, against a reading here of 27.37 °C, but the message is operator-written rather than generated, so that is a convention of the calibrating laboratory and not something the firmware does.

Action $A7 • no security gate • Published in no manufacturer document examined.

DISPlay:WINDow:TEXT:DATA12

DISPlay:WINDow:TEXT:DATA12 <string>

DISPlay:WINDow:TEXT:DATA12?

A second text-writing path alongside the documented `DISPlay:TEXT`. It carries the same action numbers, $19 and $1A, but a different parameter descriptor (descriptor index 12 rather than 11) and a different implied-node index.

From firmware analysis alone the two descriptors differ in the length they accept: the display has twelve character positions in its main field, and `DATA12` appears to be the twelve-character form.

Actions $19 / $1A • DISPlay subtree • Published in no manufacturer document examined.

L1X and L2X

L1X

L2X

Both are published, in the User's Guide, pages 148 and 222, as the way to switch between the SCPI and Keithley 181 languages. The point of interest is how they are built: they are two root-level SCPI nodes sharing a single action number, $30, and distinguished only by the implied-index field of their node records, 0 for `L1X` and 1 for `L2X`.

That is the same mechanism the parser uses for implied nodes, applied here to carry a parameter without a parameter. It is why the two commands take no argument and have no query form, and it means the Keithley language selector is reachable from the SCPI parser without the SCPI parser knowing anything about the Keithley one.

Action $30 • root level • Published by the manufacturer; see the text for where.

OUTPut:REFerence subtree

OUTPut:REFerence:GAIN?
OUTPut:REFerence:OFFSet {<value>|MIN|MAX}
OUTPut:REFerence:OFFSet:AUTO [ONCE]
OUTPut:REFerence:OFFSet:NULL [ONCE]
OUTPut:REFerence:SPAN {<value>|MIN|MAX}
OUTPut:REFerence:VALue {<value>|MIN|MAX}

OUTPut:STATe {OFF|ON}

The chart-recorder output, described in the User's Guide, pages 119 and 180. Every command in it is published there or in the compliance list except `OUTPut:REFerence:OFFSet:AUTO`, which is a second name for `OFFSet:NULL`: the two share action number $79 and take the same single keyword `ONCE`.

Confirmed on the instrument: `OUTP:REF:GAIN?` returned `+1.00000000E+03`, `OUTP:REF:SPAN?` returned `+1.00000000E-03`, `OUTP:REF:OFFS?` returned `+0.00000000E+00` and `OUTP:STAT?` returned `0`.

`VALue` has a command form only, with no query action number, so a value written to it cannot be read back.

Actions $78-$80 • no security gate • Published except for the OFFSet:AUTO spelling.

SENSe:APERture

SENSe:VOLTage:DC:APERture {<seconds>|MIN|MAX}
SENSe:RESistance:APERture {<seconds>|MIN|MAX}
SENSe:FRESistance:APERture {<seconds>|MIN|MAX}

SENSe:TEMPerature:APERture {<seconds>|MIN|MAX}

Sets the integration time in seconds. Actions $31 and $32 for every function, against $6D and $6E for the documented `NPLCycles` form. The parameter takes the unit suffixes `S` and `SEC`.

The two forms address one quantity. Confirmed on the instrument, which returned `+2.00000000E-01` for `SENS:VOLT:DC:APER?` while `SENS:VOLT:DC:NPLC?` returned `+1.00000000E+01` with `SYST:LFR?` reporting 50. Ten cycles of a 50 Hz supply are 200 ms.

Note



Because the two names set the same register, a program that sets NPLCycles and then reads APERture gets a consistent answer, and a program that sets an aperture the instrument cannot honour gets the nearest available integration time rather than an error.

Actions $31 / $32 • SENSe subtree • Published in no manufacturer document examined.

SENSe:CURRent:BIAS:ZERO:IMMediate

SENSe:CURRent:BIAS:ZERO:IMMediate

Command only, no parameters, action number $3E. Performs the bias-current zero adjustment. Published in the User's Guide's SCPI Compliance Information, page 225, though not in its command summary and not described anywhere.

What is not published is the front-panel route to it: CAL menu item `8: BIAS I`, which appears in no manufacturer menu map. Errors 630, 'Bias current DAC test failed', and 737, 'Bias current self cal failed to converge', relate to it.

Caution This is an adjustment, not a measurement. It was not exercised.



Action $3E • SENSe subtree • Published by the manufacturer; see the text for where.

SENSe:NULL and SENSe:NULL:VALue

SENSe:NULL [{OFF|ON|ONCE}]
SENSe:NULL?
SENSe:NULL:VALue {<value>|MIN|MAX}
SENSe:NULL:VALue?

Both are published, in the User's Guide's SCPI Compliance Information, page 225. The function-independent `VALue` node carries the same actions as the per-function ones, $72 and $73, so it sets the null register of whichever function is in force rather than a register of its own.

The entry is here for one anomaly. This node's implied-child field holds $00FF where every other node in the tree holds $FFFF for 'no implied child'. It is the only occurrence of that value, and its significance was not established.

Actions $72 / $73 • SENSe subtree • Published by the manufacturer; see the text for where.

SENSe:POWer:LIMit:VALue

SENSe:RESistance:POWer:LIMit:VALue {<value>|MIN|MAX}
SENSe:FRESistance:POWer:LIMit:VALue {<value>|MIN|MAX}

The User's Guide lists `POWer:LIMit[:STATe]` but gives a value node only for the voltage limit. Both exist, and both branches carry the same actions ($83 and $84 for the state, $76 and $77 for the value), so the voltage limit and the power limit are one mechanism with two names.

That shared mechanism is what errors 1000 and 1001 report. The User's Guide, page 237, describes them as raised when a later command asks for a mode or range the enabled limit does not allow, which is the same thing seen from the operator's side.

Actions $76 / $77 • SENSe subtree • The POWer:LIMit:VALue spelling is not published.

SENSe:TEMPerature:TRANsducer:FRTD:USER:COEFficients

SENSe:TEMPerature:TRANsducer:FRTD:USER:COEFficients <a4>,<b4>,<a7>,<b7>,<c7>

SENSe:TEMPerature:TRANsducer:FRTD:USER:COEFficients?

The command, its five ITS-90 deviation coefficients, the `USER` keyword of `...:FRTD:TYPE` and the front-panel `8: SPRT COEF` editor are all documented in the 34420-90091 Manual Update, pages 6 to 8. That update is not bound into the guides, which is why the facility is easy to miss.

The firmware adds two things to that account. The nodes carry actions $93 and $94, and the facility is present in all three ROM images, including revision 5.0, so it is not the later addition the update sheet's wording suggests. The instrument answered `SENS:TEMP:TRAN:FRTD:USER:COEF?` with five zeros and `SENS:TEMP:TRAN:FRTD:TYPE?` with `+85`.

Actions $93 / $94 • SENSe subtree • Published by the manufacturer; see the text for where.

SENSe:ZERO:AUTO

SENSe:ZERO:AUTO {OFF|ON|ONCE}

SENSe:ZERO:AUTO?

Controls the autozero function. Actions $35 and $36, with the enumeration `OFF`, `ON`, `ONCE`. Confirmed on the instrument, which returned `1`.

This is a different control from the equally unlisted `CALibration:ZERO:AUTO`, which carries actions $3C and $3D and lives in the calibration subtree.

Actions $35 / $36 • SENSe subtree • Published in no manufacturer document examined.

STATus:PTRansition and STATus:NTRansition

STATus:OPERation:PTRansition <mask>
STATus:OPERation:PTRansition?
STATus:OPERation:NTRansition <mask>
STATus:OPERation:NTRansition?
STATus:QUEStionable:PTRansition <mask>

STATus:QUEStionable:NTRansition <mask>

The positive- and negative-transition filters of the two status structures. Actions $29 and $2A for the positive filter and $26 and $27 for the negative one, shared by both structures. Standard SCPI, implemented, and absent from the User's Guide, which documents only the condition, event and enable registers.

A transition filter that has never been written is at its power-on default; because the User's Guide does not mention them, a program that assumes the SCPI default of all-positive-enabled should set them explicitly.

Actions $26-$2A • STATus subtree • Published in no manufacturer document examined.

SYSTem:BEEPer and SYSTem:CLICk

SYSTem:BEEPer[:IMMediate]
SYSTem:BEEPer:STATe {OFF|ON}
SYSTem:BEEPer:STATe?
SYSTem:CLICk[:IMMediate]
SYSTem:CLICk:VOLume <value>
SYSTem:CLICk:VOLume?

Sound control. `BEEPer` with no sub-node sounds the beeper once (action $39); `BEEPer:STATe` enables or disables it (actions $3A and $3B). `CLICk` is the key-click equivalent, actions $05, $06 and $07.

Confirmed on the instrument, which returned `1` for `SYST:BEEP:STAT?`. The beeper is on the earth-referenced side of the barrier (E700 on the Earth Referenced Logic sheet), so these commands travel across the isolated link before they take effect.

Actions $05-$07, $39-$3B • SYSTem subtree • Published in no manufacturer document examined.

SYSTem:LOCS, SYSTem:REMS, SYSTem:RWLS

SYSTem:LOCS
SYSTem:REMS
SYSTem:RWLS

Aliases for the documented `SYSTem:LOCal`, `SYSTem:REMote` and `SYSTem:RWLock`, carrying the same action numbers $67, $88 and $8F. The four-letter spellings are the IEEE 488.1 state names (local state, remote state, remote with lockout state), and all three were confirmed present on the instrument.

They are RS-232 commands. On the GPIB the corresponding states are entered by bus messages, and error 550, 'Command not allowed in local', or 514, 'Command allowed only with RS-232', may result. The User's Guide, page 236, gives the same restriction for the documented spellings.

Actions $67 / $88 / $8F • SYSTem subtree • Published in no manufacturer document examined.

SYSTem:LOG? and SYSTem:LOG10?

SYSTem:LOG? <value>
SYSTem:LOG10? <value>

Query-only mathematical services taking one real and returning one real. Actions $1C and $1B respectively. `LOG?` returns the natural logarithm and `LOG10?` the base-ten logarithm.

Confirmed on the instrument: `SYST:LOG? 100` returned `+4.60517000E+00` and `SYST:LOG10? 100` returned `+2.00000000E+00`.

Actions $1C / $1B • no security gate • Published in no manufacturer document examined.

SYSTem:MENU:HELP

SYSTem:MENU:HELP {OFF|ON|0|1}

SYSTem:MENU:HELP?

Enables or disables the explanatory text the front panel shows while the menu is being navigated. Actions $1F and $20; the parameter is a boolean. Confirmed on the instrument, which returned `1`.

The help strings themselves are in bank 1 of the ROM: `AT TOP OF MENU`, `MENUS`, `COMMANDS`, `PARAMETER` and `MENU BOTTOM`.

Actions $1F / $20 • no security gate • Published in no manufacturer document examined.

TEST:TEMPerature?

TEST:TEMPerature? <real>,<integer>

A root-level query with action number $2C, taking a real followed by an integer and returning six items. Sent with no parameter it returns error -109, 'Missing parameter'; sent with one integer it returns a syntax error. The values it accepts were not determined.

From firmware analysis alone this is a transducer conversion service: the real is a transducer reading and the integer selects the transducer, and the query returns the corresponding temperature without changing the measurement configuration. Its parameter descriptors are the same descriptors the temperature measurement commands use.

Action $2C • root level • Published in no manufacturer document examined.

TRACe:FEED and TRACe:POINts?

TRACe:FEED RDG_STORE,{"CALC"|""}

TRACe:FEED? RDG_STORE

TRACe:POINts? RDG_STORE

Aliases of the documented `DATA:FEED` and `DATA:POINts?`, carrying the same actions $55, $56 and $68. SCPI defines the reading-memory commands under `TRACe` and permits `DATA` as an alternative; this instrument implements both and the User's Guide documents only the second.

Actions $55 / $56 / $68 • root level • Published in no manufacturer document examined.

12

SCPI Keyword Table

Keyword Table

The complete keyword table from bank 3 of the revision 10.0 image, in ROM order. The same 182 keywords appear in revisions 9.0 and 5.0, at $B744 in revision 5.0 and $B74C in the other two.

Keywords are printed with the short form in upper case and the remainder of the long form in lower case, the convention the User's Guide describes on page 208. A keyword shown entirely in upper case has no long form.

The table serves two purposes at once. A keyword may be a command mnemonic, a discrete parameter value, or both, which is why `MINimum`, `AUTO` and `ONCE` sit in the same table as `CALibration` and `TRIGger`. Which of them a particular command accepts is fixed by the enumeration list attached to its parameter descriptor, and is given for every command in Chapter 11.

Three entries are of note. `_TRG` cannot be reached through the SCPI lexer at all, as Chapter 11 records. `DATA12` belongs to a command that appears in no manufacturer document. `SENSE1`, `SENS1`, `SENSE2` and `SENS2` are separate keywords rather than a numeric suffix on `SENSe`, which is why the whole measurement subtree is repeated under each of the six spellings of its root.

Table 12-1. SCPI Keyword Table, 182 Entries

Keyword	Keyword	Keyword	Keyword	Keyword	Keyword
SENS	CLICk	APERture	FETCh	LFRequency	FRONt
SENSe	BEEPer	NPLCycles	DATA12	FLASh	RDG_STORE
CAL	LOCS	RESolution	DATA	COUNt	BOTH
CALibration	LOCal	UPPer	CLEar	ICURrent	DIGital
WAI	RWLS	RANGe	TEXT	ALL	ANALog
OPC	RWLock	LIMit	WINDow	ABORt	FAST
_TRG	REMS	POWer	DISPlay	SOFT	MEDium
TRG	REMote	OCOMpensated	AMBient	NORMal	SLOW
PSC	LANGuage	BIAS	POKE	INTernal	RS232
STB	INTerface	CURRent	PEEK	FIXed	HPIB
SRE	PARity	SENSe2	DIAGnostic	FAR	ODD
ESR	BAUD	SENS2	TEMPerature	F	NONE
ESE	TRANsmit	SENSe1	FRESistance	CEL	EVEN
TST	RECeive	SENS1	RESistance	C	AVERage
RST	SERial	SAMPle	RATio	T	SCALe
IDN	COMMunicate	TERMinals	DIFFerence	S	EXTernal
CLS	LOG10	ROUTe	DC	R	IMMediate
UNIT	LOG	READ	VOLTage	N	BUS
SOURce	SYSTem	SPAN	SCALar	K	ON
DELay	PRESet	NULL	CONFigure	J	OFF
SEQuence	PTRansition	REFerence	OFFSet	E	ONCE
TRIGger	NTRansition	MEASure	PTPeak	B	USER
POINts	ENABle	L2X	SDEViation	FRTD	DEFault
FEED	EVENt	L1X	FUNCtion	THERmistor	AUTO
TRACe	CONDition	TYPE	CALCulate	TCouple	INFinity
TEST	OPERation	PRECharge	VALue	GAIN	MINimum
VERSion	QUEStionable	RESPonse	STRing	ZERO	MAXimum
HELP	STATus	LPASs	STATe	FRONT2	
MENU	COEFficients	FILTer	CODE	FRON2	
ERRor	RJUNction	INPut	SECure	FRONT1	
VOLume	TRANsducer	INITiate	OUTPut	FRON1	

The single-letter entries `B`, `E`, `J`, `K`, `N`, `R`, `S`, `T`, `C` and `F` are thermocouple types and temperature units, not commands.

13

Quick Reference

Key Addresses

Table 13-1. Key Addresses, Revision 10.0

Address	What it is
$2080	Reset entry, all banks
$20D4	Call to the bank's initialisation entry (displacement patched per bank)
$20EB	Call to the bank's main loop (displacement patched per bank)
$2120	Unexpected-interrupt trap; dumps 15 words to $1488 and resets
$2132/$2138/$213E/$2144	Inter-bank call trampolines for banks 0/1/2/3
$2148	Common trampoline body
$2164	Operand holding this bank's vector table base
$216D	Inter-bank return stub
$21AC	Front-panel / gate-array interrupt handler (INT07)
$259A	Timer overflow handler; software time base in R3Eh:R40h
$25E0	Line-frequency sense handler
$35AE	Bare RET used as the unexported-entry stub
$35B4	First address that differs between banks
$1488	Crash buffer in RAM, 15 words
$17F8	Inter-bank transfer stack, grows upward
$1FF8	Gate-array status and control
$1FF9	Front-panel data register
$1FFB	Front-panel interrupt acknowledge

Reading the Instrument Without Risk

Commands that read state and change nothing, useful when taking an instrument in for service. Eight of the eleven are described in the User's Guide, pages 115 and 197 to 198; the value of the list is the selection, not the descriptions.

```
*IDN?                                revisions: main, I/O, front panel
SYSTem:ERRor?                        oldest queued error
CALibration:COUNt?                   number of calibration steps performed
CALibration:STRing?                  stored calibration message
CALibration:SECure:STATe?            1 if secured
SYSTem:LFRequency?                   sensed line frequency
DIAGnostic:TEMPerature:AMBient?      internal temperature, degrees C
SENSe:FUNCtion?                      present function
ROUTe:TERMinals?                     selected input channel
STATus:QUEStionable:CONDition?       questionable-data condition register
DIAGnostic:PEEK? 1,5256,1            first word of the crash buffer
```

Caution



Everything else in the `CALibration` subtree either changes the security state or performs an adjustment. Read Chapter 7 before sending any of it.

Bank Map

Table 13-2. What Is in Each Bank

Bank	PORT1 7:6	Contents	Evidence
0	00	A/D control, calibration arithmetic, measurement maths	Strings `CAL VALUE`, `A/D RUNDOWN`, `PRECH DAC`, `INJ CURRENT`
1	01	Front panel: display formatting, menu, keyboard	Menu strings, `MENUS`, `PARAMETER`, the function-generator residue
2	10	Message text, function names, language emulation	The error table, `VOLT:DC`, `FRON1`, the EEPROM defaults
3	11	Executive and SCPI parser; the boot bank	Keyword table, command nodes, `@(#) SCPI Version 2.22`

Document History

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